

第三讲

**简单截面数据
的经济计量方法
(二)**

总结：高斯-马尔可夫定理

- 在高斯马尔可夫假定1-假定4下，参数的OLS估计是最优线性无偏估计 (BLUE: Best Linear unbiased estimator)
 - 没有其他的估计会比OLS估计更好！！
- 问：在社会经济问题研究中，高斯-马尔可夫假定哪些不易满足？
 - 假定： $E(u | x) = E(u) = 0$ 不易满足，导致：
 - 估计量是有偏估计和不一致的
 - 应用中可能出现OLS估计值的含义与实际背离的问题

- 例2.12(第2章): 学生免费午餐比例对10年级学生数学成绩的影响(meap93.txt)
 - 408所高中的数据: lnmprg:参与午餐资助计划的学生百分比; Math10:10年级学生数学成绩通过的百分比。建立一元线性回归模型进行实证研究 $math10 = \beta_0 + \beta_1 lnmprg + u$

```
> Data<-read.table(file="meap93.txt",header=TRUE,sep="")
> ypred<-lm(MATH10~LNCHPRG,data=Data)
> summary(ypred)

Call:
lm(formula = MATH10 ~ LNCHPRG, data = Data)

Residuals:
    Min       1Q   Median       3Q      Max
-24.386  -5.979  -1.207   4.865  45.845

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  32.14271    0.99758   32.221  <2e-16 ***
LNCHPRG      -0.31886    0.03484   -9.152  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 9.566 on 406 degrees of freedom
Multiple R-squared:  0.171,    Adjusted R-squared:  0.169
F-statistic: 83.77 on 1 and 406 DF,  p-value: < 2.2e-16
```

. reg MATH10 LNCHPRG

Source	SS	df	MS	Number of obs	=	408
Model	7665.26605	1	7665.26605	F(1, 406)	=	83.77
Residual	37151.9147	406	91.5071792	Prob > F	=	0.0000
Total	44817.1808	407	110.115923	R-squared	=	0.1710
				Adj R-squared	=	0.1690
				Root MSE	=	9.5659

MATH10	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
LNCHPRG	-0.3188643	.0348393	-9.15	0.000	-.3873523	-.2503763
_cons	32.14271	.9975824	32.22	0.000	30.18164	34.10378

lnmprg通常与math10为正向关系; 怀疑u中包含了x高度相关的因素, 如学生特征(所在地区学生家庭的贫困率)、学校资源等

- 高斯马尔可夫假定SLR3不满足的重要原因之一: 存在自变量遗漏
- 处理方法: 对可测度的变量, 总体模型选择多元线性回归模型

主要内容

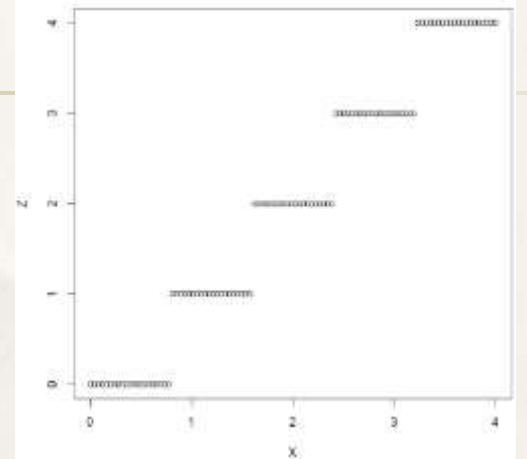
- 不满足假定3情况下的多元线性回归解决策略
 - 多元线性回归模型的一般问题
 - 如何验证所设定的多元模型是恰当的
 - 模型设定的问题
 - 模型的误设和诊断

多元线性回归模型的一般问题

- 多元模型的必要性
- 多元模型参数的含义
- 高斯-马尔可夫假定与多元线性回归模型参数的性质
- 变量重要性的对比

多元线性回归模型

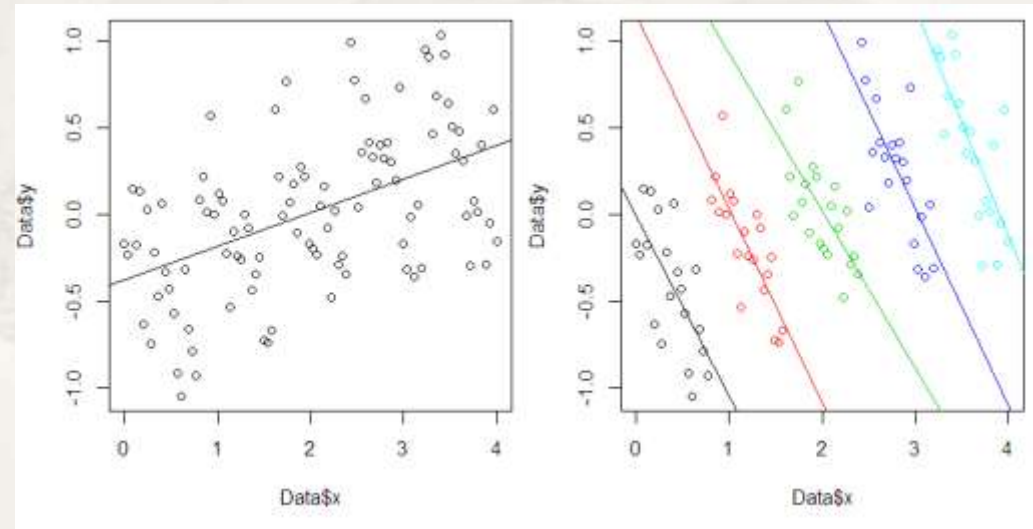
- 多元线性回归的必要性：**假定3**明显不满足下对影响效应的错误估计（通过数据模拟进行研究）



```
X<-seq(from=0,to=4,length=100) #100个数
Z<-rep(0:4,each=20) #100个数, Z的变化较X要大
plot(X,Z)
Y<-(-X)+Z+rnorm(100,0,0.25) #Y是X和Z的线性函数加上随机误差项(均值为0, 标准差为0.25)
Data<-data.frame(x=X,y=Y,z=Z)
plot(Data$x,Data$y,pch=1)
(M0<-lm(y~x,data=Data))
abline(M0) ##X,Y正相关
plot(Data$x,Data$y,pch=1,col=Data$z+1)
for(i in 0:4){
  T.Data<-subset(Data,Data$z==i)
  M0<-lm(y~x,data=T.Data)
  abline(M0,col=i+1) #控制了Z后, X,Y呈负相关
}
(lm(y~x+z,data=Data))
```

```
Coefficients:
(Intercept)      x
   -0.4153      0.1937
```

```
Coefficients:
(Intercept)      x      z
   -0.03309   -0.98702    0.98309
```



多元线性回归模型

- 总体模型：多元线性回归模型

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \dots + \beta_k x_k + u.$$

- 参数估计前提：经典线性模型假定

- 高斯马尔可夫假定1、2不变
- 假定3(零条件均值)扩展为：

$$E(u|x) = 0. \quad \longrightarrow \quad E(u|x_1, x_2, \dots, x_k) = 0.$$

- 意味：误差项 u 与任何解释变量及其任何函数都无关
- 如果 x_j 与 u 不相关，则称 x_j 为外生解释变量(exogenous explanatory variables)
- 如果 x_j 与 u 相关，则称 x_j 为内生解释变量(endogenous explanatory variables)

多元线性回归模型的参数估计

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \dots + \beta_k x_k + u.$$

- 从假定3出发的矩法估计 $E(u) = 0$ $E(x_j u) = 0, j = 1, 2, \dots, k.$

$$\begin{aligned}\sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \dots - \hat{\beta}_k x_{ik}) &= 0 \\ \sum_{i=1}^n x_{i1}(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \dots - \hat{\beta}_k x_{ik}) &= 0 \\ \sum_{i=1}^n x_{i2}(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \dots - \hat{\beta}_k x_{ik}) &= 0 \\ \vdots \\ \sum_{i=1}^n x_{ik}(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \dots - \hat{\beta}_k x_{ik}) &= 0.\end{aligned}$$

多元中OLS的一阶条件,同最小二乘估计

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2 + \dots + \hat{\beta}_k x_k$$

最小二乘估计:

$$\sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \dots - \hat{\beta}_k x_{ik})^2.$$

$$\begin{aligned}\sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \dots - \hat{\beta}_k x_{ik}) &= 0 \\ \sum_{i=1}^n x_{i1}(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \dots - \hat{\beta}_k x_{ik}) &= 0 \\ \sum_{i=1}^n x_{i2}(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \dots - \hat{\beta}_k x_{ik}) &= 0 \\ \vdots \\ \sum_{i=1}^n x_{ik}(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \dots - \hat{\beta}_k x_{ik}) &= 0.\end{aligned}$$

多元线性回归模型参数的含义

- 在其他情况不变的条件下， x_i 增加一个单位引起的 y 的变动等于 β_i

$$\Delta \hat{y} = \hat{\beta}_1 \Delta x_1 + \hat{\beta}_2 \Delta x_2 + \dots + \hat{\beta}_k \Delta x_k$$



误差项和其他变量保持不变时

$$\Delta \hat{y} = \hat{\beta}_1 \Delta x_1$$

- 注意两个假设条件：
 - 模型中其他变量不变
 - u 中其他因素不变

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \dots + \beta_k x_k + u$$

■ 对其他不变的理解：扣除自变量对因变量可解释部分中也可被其他变量解释的部分

例如：教育回报率问题

$$wage = \beta_0 + \beta_1 edu + \beta_2 exper + u$$

```
> lgWAGE_lm1<-lm(log(WAGE)~EDUC+EXPER,data=Data)
> summary(lgWAGE_lm1)
```

```
Call:
lm(formula = log(WAGE) ~ EDUC + EXPER, data = Data)
```

```
Residuals:
    Min       1Q   Median       3Q      Max
-2.05800 -0.30136 -0.04539  0.30601  1.44425
```

```
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.216854   0.108595   1.997   0.0464 *
EDUC         0.097936   0.007622  12.848 < 2e-16 ***
EXPER        0.010347   0.001555   6.653 7.24e-11 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 0.4614 on 523 degrees of freedom
Multiple R-squared:  0.2493,    Adjusted R-squared:  0.2465
F-statistic: 86.86 on 2 and 523 DF,  p-value: < 2.2e-16
```

```
> educ_lm<-lm(EDUC~EXPER,data=Data) #辅助回归
> lm(log(WAGE)~educ_lm$residuals,data=Data)
```

```
Call:
lm(formula = log(WAGE) ~ educ_lm$residuals, data = Data)
```

```
Coefficients:
(Intercept)  educ_lm$residuals
    1.62327         0.09794
```

```
. reg LNWAGE EDUC EXPER
```

Source	SS	df	MS	Number of obs	=	526
Model	36.9850401	2	18.4925201	F(2, 523)	=	86.86
Residual	111.34471	523	.212896195	Prob > F	=	0.0000
Total	148.32975	525	.282532857	R-squared	=	0.2493
				Adj R-squared	=	0.2465
				Root MSE	=	0.4614

LNWAGE	Coef.	Std. Err.	t	P> t
EDUC	0.0979356	0.0076224	12.85	0.000
EXPER	0.0103469	0.0015551	6.65	0.000
_cons	0.2168544	0.108595	2.00	0.046

Source	SS	df	MS	Number of obs	=	526
Model	361.182951	1	361.182951	F(1, 524)	=	51.65
Residual	3664.24671	524	6.99283723	Prob > F	=	0.0000
Total	4025.42966	525	7.66748506	R-squared	=	0.0897
				Adj R-squared	=	0.0880
				Root MSE	=	2.6444

EDUC	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]
EXPER	-0.0611132	0.0085035	-7.19	0.000	-0.0778184 -0.044408
_cons	13.60271	0.1850242	73.52	0.000	13.23923 13.96619

```
. predict e1, residual
```

```
. reg LNWAGE e1
```

Source	SS	df	MS	Number of obs	=	526
Model	35.1451704	1	35.1451704	F(1, 524)	=	162.71
Residual	113.18458	524	.216001106	Prob > F	=	0.0000
Total	148.32975	525	.282532857	R-squared	=	0.2369
				Adj R-squared	=	0.2355
				Root MSE	=	0.46476

LNWAGE	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]
e1	0.0979356	0.0076778	12.76	0.000	0.0828526 0.1130186
_cons	1.623268	0.0202645	80.10	0.000	1.583459 1.663078



■ 规范表述，以二元为例：

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2$$

- **辅助回归：**利用样本数据做 x_1 对 x_2 的简单回归得到残差 $\hat{r}_1$ ，它是 x_1 中与 x_2 不相关的部分
- **主回归：**做 y 对辅助回归残差的回归得到 β_1

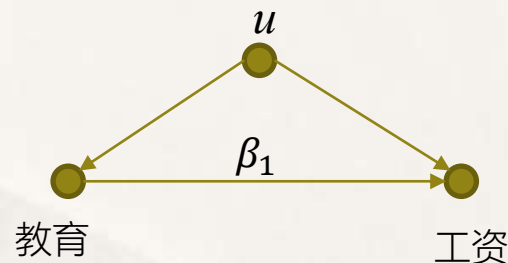
$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})y_i}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

$$\hat{\beta}_1 = \left(\sum_{i=1}^n \hat{r}_{i1} y_i \right) / \left(\sum_{i=1}^n \hat{r}_{i1}^2 \right),$$

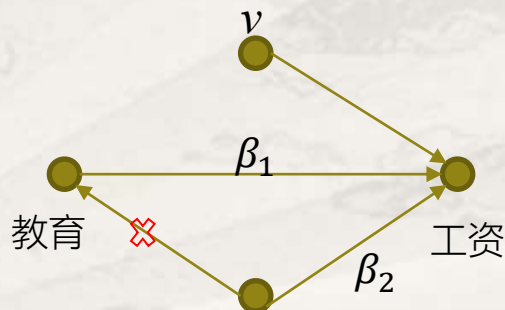
- β_1 体现了 x_1 对 y 的“净解释能力”（扣除了可被 x_2 解释的部分）

$$wage = \beta_0 + \beta_1 edu + u$$



$$u = \beta_2 exper + v$$

$$wage = \beta_0 + \beta_1 edu + \beta_2 exper + v$$



经验

β_1 是扣除了经验的干扰性关联效应后得到的教育的因果关联效应的估计

■ 证明:

$$\hat{\beta}_1 = \left(\sum_{i=1}^n \hat{r}_{i1} y_i \right) / \left(\sum_{i=1}^n \hat{r}_{i1}^2 \right),$$

辅助回归
的残差

■ 将 x_1 对 $x_2, \dots, x_k$ 做辅助回归, 有: $x_{i1} = \hat{x}_{i1} + \hat{r}_{i1}, i = 1, \dots, n.$

■ 代入原回归方程OLS的一阶条件方程, 有:

$$\sum_{i=1}^n (\hat{x}_{i1} + \hat{r}_{i1})(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \dots - \hat{\beta}_k x_{ik}) = 0.$$

原模型残差

$\hat{x}_{i1}$ 是自变量的线性函数, 根据原模型的残差性质:

$$\text{Cov}(\hat{x}_1, \hat{u}) = 0, \text{ 有: } \sum_{i=1}^n \hat{x}_{i1} \hat{u}_i = 0$$

$$\sum_{i=1}^n \hat{r}_{i1}(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \dots - \hat{\beta}_k x_{ik}) = 0.$$

$\therefore \hat{r}_{i1}$ 是残差, 与 $x_j (j=2, \dots, k)$ 等无关, 有: $\sum_{i=1}^n \hat{r}_{i1} x_{ij} = 0$

$$\sum_{i=1}^n \hat{r}_{i1}(y_i - \hat{\beta}_1 x_{i1}) = 0.$$

$$\sum_{i=1}^n \hat{r}_{i1}(y_i - \hat{\beta}_1 \hat{r}_{i1}) = 0.$$

$$\therefore \sum_{i=1}^n \hat{r}_{i1}(y_i - \hat{\beta}_1 \hat{x}_{i1} - \hat{\beta}_1 \hat{r}_{i1}) = 0, \sum_{i=1}^n \hat{\beta}_1 \hat{x}_{i1} \hat{r}_{i1} = 0$$

$$\begin{aligned} \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \dots - \hat{\beta}_k x_{ik}) &= 0 \\ \sum_{i=1}^n x_{i1}(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \dots - \hat{\beta}_k x_{ik}) &= 0 \\ \sum_{i=1}^n x_{i2}(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \dots - \hat{\beta}_k x_{ik}) &= 0 \\ &\vdots \\ \sum_{i=1}^n x_{ik}(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \dots - \hat{\beta}_k x_{ik}) &= 0. \end{aligned}$$

- 问：如何保证多元模型中OLS估计量的无偏性？

- 假定3

- 证明： $E(\hat{\beta}_j) = \beta_j, j = 0, 1, \dots, k$ ，以 β_1 的证明为例

- β_1 可通过 y 对辅助回归的残差 $\hat{r}_1$ (x_1 对其他自变量回归所得) 的一元线性回归得到，参照一元模型：

$$\hat{\beta}_1 = \beta_1 + \frac{\sum_{i=1}^n (x_i - \bar{x})u_i}{s_x^2}$$

- 有：

$$\hat{\beta}_1 = \beta_1 + \left(\sum_{i=1}^n \hat{r}_{i1} u_i \right) / \left(\sum_{i=1}^n \hat{r}_{i1}^2 \right).$$



$$E(u|x_1, x_2, \dots, x_k) = 0.$$

$$\begin{aligned} E(\hat{\beta}_1|X) &= \beta_1 + \left(\sum_{i=1}^n \hat{r}_{i1} E(u_i|X) \right) / \left(\sum_{i=1}^n \hat{r}_{i1}^2 \right) \\ &= \beta_1 + \left(\sum_{i=1}^n \hat{r}_{i1} \cdot 0 \right) / \left(\sum_{i=1}^n \hat{r}_{i1}^2 \right) = \beta_1, \end{aligned}$$

■ 问：如何度量多元模型中OLS估计量的方差？（第3章）

■ 高斯-马尔可夫假定4(同方差性)扩展为： $\text{Var}(u|x_1, \dots, x_k) = \sigma^2$.

■ 在假定4下，估计量的方差为：

$$\hat{\beta}_1 = \beta_1 + \left(\sum_{i=1}^n \hat{r}_{i1} u_i \right) / \left(\sum_{i=1}^n \hat{r}_{i1}^2 \right).$$

$$\begin{aligned} \text{Var}(\hat{\beta}_1|X) &= \left(\sum_{i=1}^n \hat{r}_{i1}^2 \text{Var}(u_i|X) \right) / \left(\sum_{i=1}^n \hat{r}_{i1}^2 \right)^2 \\ &= \left(\sum_{i=1}^n \hat{r}_{i1}^2 \sigma^2 \right) / \left(\sum_{i=1}^n \hat{r}_{i1}^2 \right)^2 = \sigma^2 / \left(\sum_{i=1}^n \hat{r}_{i1}^2 \right). \end{aligned}$$

分母为辅助回归的残差平方和，
对于辅助回归有： $SST_1 = SSE_1 + SSR_1$

有： $SSR_1 = (1 - R_1^2)SST_1$

$$\text{Var}(\hat{\beta}_j) = \frac{\sigma^2}{SST_j(1 - R_j^2)}$$

$$SST_j = \sum_{i=1}^n (x_{ij} - \bar{x}_j)^2$$

$$\begin{aligned} R_1^2 &= \frac{SSE_1}{SST_1} = 1 - \frac{SSR_1}{SST_1} \\ \hat{\sigma}^2 &= \frac{\sum_{i=1}^n \hat{u}_i^2}{n - k - 1} = \frac{SSR}{n - k - 1} \end{aligned}$$

自由度为什么是这个数？

$$E(u) = 0$$

$$E(x_j u) = 0, j = 1, 2, \dots, k.$$

- 进一步： $\hat{\beta}_j$ 的抽样分布

- 经典线性模型假定 (附加假定) 下：

$$y|\mathbf{x} \sim \text{Normal}(\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_k x_k, \sigma^2)$$

$$\hat{\beta}_j \sim \text{Normal}[\beta_j, \text{Var}(\hat{\beta}_j)].$$

- 若无法满足附加假定，大样本下OLS估计量仍具有渐近性

- 构造t统计量进行回归系数的显著性检验：

$$\hat{\beta}_j \sim \text{Normal}[\beta_j, \text{Var}(\hat{\beta}_j)].$$

$$(\hat{\beta}_j - \beta_j) / \text{sd}(\hat{\beta}_j) \sim \text{Normal}(0, 1)$$

$$(\hat{\beta}_j - \beta_j) / \text{se}(\hat{\beta}_j) \sim t_{n-k-1}.$$

$$t_{\hat{\beta}_j} \equiv \hat{\beta}_j / \text{se}(\hat{\beta}_j)$$

- 得到系数的置信区间

例：教育回报率问题

```
> lgWAGE_lm1<-lm(log(WAGE)~EDUC+EXPER,data=Data)
> summary(lgWAGE_lm1)
```

```
Call:
lm(formula = log(WAGE) ~ EDUC + EXPER, data = Data)
```

Residuals:

```
      Min       1Q   Median       3Q      Max
-2.05800 -0.30136 -0.04539  0.30601  1.44425
```

Coefficients:

```
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 0.216854   0.108595   1.997   0.0464 *
EDUC         0.097936   0.007622  12.848 < 2e-16 ***
EXPER        0.010347   0.001555   6.653 7.24e-11 ***
```

. reg LNWAGE EDU EXPER

Source	SS	df	MS	Number of obs	=	526
Model	36.9850401	2	18.4925201	F(2, 523)	=	86.86
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Total	148.32975	525	.282532857	R-squared	=	0.2493
				Adj R-squared	=	0.2465
				Root MSE	=	.46141

LNWAGE	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
EDUC	.0979356	.0076224	12.85	0.000	.0829613	.1129099
EXPER	.0103469	.0015551	6.65	0.000	.0072919	.013402
_cons	.2168544	.108595	2.00	0.046	.0035183	.4301904

- 再看估计量的方差

$$\text{Var}(\hat{\beta}_j) = \frac{\sigma^2}{\text{SST}_j(1 - R_j^2)}$$

$$\text{SST}_j = \sum_{i=1}^n (x_{ij} - \bar{x}_j)^2$$

- 多重共线性的度量：容忍度

它是什么？也影响估计量的方差

$$1 - R_j^2$$

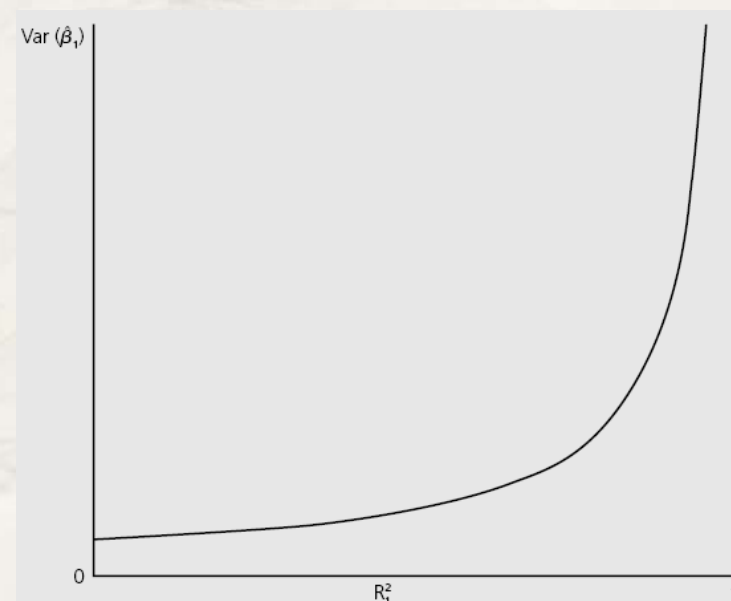
x_j 对其他自变量回归的拟合优度

- 容忍度反映了 x_j 中其他自变量无法解释的部分，越小说明 x_j 与其他自变量间的共线性越强(第3章)

- x_j 与其他解释变量的共线性越大(容忍度越小)，估计量的方差越大，估计值波动性越大，一次估计值越可能不可靠(如图)

- 为确保分母不等于0，需增加假设

共线性对斜率参数方差的影响



$$\text{Var}(\hat{\beta}_j) = \frac{\sigma^2}{\text{SST}_j(1 - R_j^2)}$$

- 增加高斯-马尔科夫假定5: 不存在完全共线性
 - 如果回归方程中的某个自变量是其他自变量的一个线性函数, 则该模型存在完全共线性

$$x_3 = x_1 + x_2 \quad x_2 = x_1^2$$

哪个存在完全共线性?

- 该假定不成立的情况:
 - 一个自变量是另一个自变量的常数倍
 - 一个自变量恰好可表示成其他自变量的线性函数
- 多重共线性的诊断:
 - 计算各解释变量的容忍度: $1 - R_j^2$
 - 计算各解释变量的方差膨胀因子 (variance inflation factor, VIF): $VIF_j = \frac{1}{1 - R_j^2}$
 - 容忍度越小或VIF越大, x_j 与其他解释变量的多重共线性越高
 - 通常VIF大于10即容忍度小于0.1认为多重共线性较强

■ 例：教育回报率研究

```
> lgWAGE_lm1<-lm(log(WAGE)~EDUC+EXPER,data=Data)
> 1/vif(lgWAGE_lm1)
      EDUC      EXPER
0.9102747 0.9102747
> lgWAGE_lm1<-lm(log(WAGE)~EDUC+EXPER+TENURE,data=Data)
> 1/vif(lgWAGE_lm1)
      EDUC      EXPER      TENURE
0.8986577 0.6767650 0.7411275
```

. reg LNWAGE EDU EXPER

Source	SS	df	MS	Number of obs	=	526
Model	36.9850401	2	18.4925201	F(2, 523)	=	86.86
Residual	111.34471	523	.212896195	Prob > F	=	0.0000
				R-squared	=	0.2493
				Adj R-squared	=	0.2465
Total	148.32975	525	.282532857	Root MSE	=	.46141

LNWAGE	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]
EDUC	.0979356	.0076224	12.85	0.000	.0829613 .1129099
EXPER	.0103469	.0015551	6.65	0.000	.0072919 .013402
_cons	.2168544	.108595	2.00	0.046	.0035183 .4301904

. vif

Variable	VIF	1/VIF
EDUC	1.10	0.910275
EXPER	1.10	0.910275
Mean VIF	1.10	

容忍度

■ EDUC和EXPER的多重共线性很弱

■ EDUC和EXPER+TENURE的共线性略高于仅与EXPER的共线性

■ 整体上多重共线性很弱

. reg LNWAGE EDU EXPER TENURE

Source	SS	df	MS	Number of obs	=	526
Model	46.874178	3	15.624726	F(3, 522)	=	80.39
Residual	101.455572	522	.194359333	Prob > F	=	0.0000
				R-squared	=	0.3160
				Adj R-squared	=	0.3121
Total	148.32975	525	.282532857	Root MSE	=	.44086

LNWAGE	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]
EDUC	.092029	.0073299	12.56	0.000	.0776292 .1064288
EXPER	.0041211	.0017233	2.39	0.017	.0007357 .0075065
TENURE	.0220672	.0030936	7.13	0.000	.0159897 .0281448
_cons	.2843595	.1041904	2.73	0.007	.0796756 .4890435

. vif

Variable	VIF	1/VIF
EXPER	1.48	0.676765
TENURE	1.35	0.741127
EDUC	1.11	0.898658
Mean VIF	1.31	

容忍度

多重共线性带来的问题

- 多重共线性对因果效应估计的影响

- $R_j^2 \rightarrow 1$ 并不违背假设3

- 尽管高共线性会导致 $\text{Var}(\hat{\beta}_j)$ 增大，但小样本等也会导致，共线性和样本量哪个更需关注是值得考虑的

$$\text{Var}(\hat{\beta}_j) = \frac{\sigma^2}{\text{SST}_j(1 - R_j^2)}$$

- 共线性是否对关注的自变量(x_1)产生影响

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + u$$

- 例：考察教育回报率时，控制EXPER，TENURE等

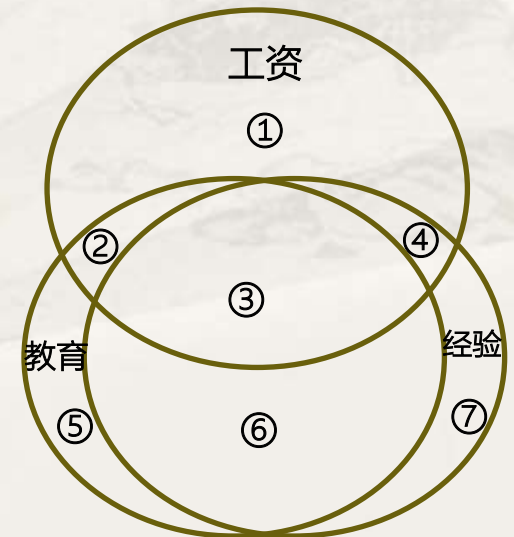
. corr EDU EXPER TENURE (obs=526)				
	EDUC	EXPER	TENURE	
EDUC	1.0000			
EXPER	-0.2995	1.0000		
TENURE	-0.0562	0.4993	1.0000	

- 如果建模的目的仅是考察自变量 x_1 的偏效应，则不必太顾忌 x_2 、 x_3 的相关，并没有对 $\text{Var}(\hat{\beta}_1)$ 产生直接影响

- 自变量的高度共线性是有问题的(如图)

- 估计值小、估计量方差大，导致统计不显著但实际显著

$$t_{\hat{\beta}_j} \equiv \hat{\beta}_j / \text{se}(\hat{\beta}_j)$$



多元线性回归模型

- 假定总结：
 - 经典线性模型假定包括：
 - 高斯-马尔可夫假定1-5
 - 附加假定(误差项正态分布)
- 经典线性模型假定下，OLS估计是最优线性无偏估计(BLUE)

自变量重要性的对比

- 可基于**标准化回归系数**比较自变量对因变量的重要程度(第6章)

$$y_i = \hat{\beta}_0 + \hat{\beta}_1 x_{i1} + \hat{\beta}_2 x_{i2} + \dots + \hat{\beta}_k x_{ik} + \hat{u}_i$$

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x = (\bar{y} - \hat{\beta}_1 \bar{x}) + \hat{\beta}_1 x = \bar{y} + \hat{\beta}_1 (x - \bar{x})$$

$x = \bar{x}$ 时有: $\hat{y} = \bar{y}$, 样本均值点落在回归线上

↓ 去减均值方程

$$y_i - \bar{y} = \hat{\beta}_1 (x_{i1} - \bar{x}_1) + \hat{\beta}_2 (x_{i2} - \bar{x}_2) + \dots + \hat{\beta}_k (x_{ik} - \bar{x}_k) + \hat{u}_i$$

↓

$$(y_i - \bar{y})/\hat{\sigma}_y = (\hat{\sigma}_1/\hat{\sigma}_y)\hat{\beta}_1[(x_{i1} - \bar{x}_1)/\hat{\sigma}_1] + \dots + (\hat{\sigma}_k/\hat{\sigma}_y)\hat{\beta}_k[(x_{ik} - \bar{x}_k)/\hat{\sigma}_k] + (\hat{u}_i/\hat{\sigma}_y).$$

$$\hat{b}_j = (\hat{\sigma}_j/\hat{\sigma}_y)\hat{\beta}_j \text{ for } j = 1, \dots, k.$$

↓

$$z_y = \hat{b}_1 z_1 + \hat{b}_2 z_2 \dots + \hat{b}_k z_k + \text{error}$$

标准化回归系数是自变量增加一个标准差单位, 对因变量标准差单位产生的影响。故可直接依据 $\hat{b}_j$ 进行变量重要性对比

- 进一步, 问: 如何知道重要性的**差异**是否**显著**呢?

■ 检验重要性 **差异的显著性**：参数的线性组合假设检验（第4章）

- 例如：多上一年学与多一年工作经验，对工资的影响是否有显著差异？

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	0.216854	0.108595	1.997	0.0464	*
EDUC	0.097936	0.007622	12.848	< 2e-16	***
EXPER	0.010347	0.001555	6.653	7.24e-11	***

. reg LNWAGE EDU EXPER

Source	SS	df	MS	Number of obs	=	526
Model	36.9850401	2	18.4925201	F(2, 523)	=	86.86
Residual	111.34471	523	.212896195	Prob > F	=	0.0000
Total	148.32975	525	.282532857	R-squared	=	0.2493
				Adj R-squared	=	0.2465
				Root MSE	=	.46141

LNWAGE	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]
EDUC	.0979356	.0076224	12.85	0.000	.0829613 .1129099
EXPER	.0103469	.0015551	6.65	0.000	.0072919 .013402
_cons	.2168544	.108595	2.00	0.046	.0035183 .4301904

$$\log(\text{wage}) = \beta_0 + \beta_1 \text{educ} + \beta_2 \text{exper}$$

$$H_0: \beta_1 - \beta_2 = 0$$

$$t = \frac{\hat{\beta}_1 - \hat{\beta}_2}{\text{se}(\hat{\beta}_1 - \hat{\beta}_2)}$$

- 如何得到检验统计量的标准误？

$$\text{se}(\hat{\beta}_1 - \hat{\beta}_2) \stackrel{?}{=} \text{se}(\hat{\beta}_1) - \text{se}(\hat{\beta}_2)$$

- 因标准误可能是负的，不合理！

- 解决方法：

$$\text{Var}(\hat{\beta}_1 - \hat{\beta}_2) = \text{Var}(\hat{\beta}_1) + \text{Var}(\hat{\beta}_2) - 2 \text{Cov}(\hat{\beta}_1, \hat{\beta}_2).$$

$$\text{se}(\hat{\beta}_1 - \hat{\beta}_2) = \{[\text{se}(\hat{\beta}_1)]^2 + [\text{se}(\hat{\beta}_2)]^2 - 2s_{12}\}^{1/2}$$

■ 检验重要性差异的显著性:

■ 参数的线性组合假设检验:

$$\log(\text{wage}) = \beta_0 + \beta_1 \text{educ} + \beta_2 \text{exper}$$

$$\theta_1 = \beta_1 - \beta_2.$$

$$H_0: \theta_1 = 0$$

$$\log(\text{wage}) = \beta_0 + \beta_1 \text{educ} + \beta_2 \text{exper}$$

$$\log(\text{wage}) = \beta_0 + (\theta_1 + \beta_2) \text{educ} + \beta_2 \text{exper}$$

$$= \beta_0 + \theta_1 \text{educ} + \beta_2 (\text{educ} + \text{exper})$$

```
Data$EDUEXP <- with(Data, EDUC + EXPER)
lgWAGE_lm2 <- lm(log(WAGE) ~ EDUC + EDUEXP, data = Data)
```

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.216854	0.108595	1.997	0.0464 *
EDUC	0.087589	0.007309	11.984	< 2e-16 ***
EDUEXP	0.010347	0.001555	6.653	7.24e-11 ***

```
. gen eduexp=EDUC+EXPER
```

```
. reg LNWAGE EDU eduexp
```

Source	SS	df	MS	Number of obs	=	526
Model	36.9850401	2	18.4925201	F(2, 523)	=	86.86
Residual	111.34471	523	.212896195	Prob > F	=	0.0000
				R-squared	=	0.2493
				Adj R-squared	=	0.2465
Total	148.32975	525	.282532857	Root MSE	=	.46141

LNWAGE	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]
EDUC	.0875886	.0073088	11.98	0.000	.0732305 .1019468
eduexp	.0103469	.0015551	6.65	0.000	.0072919 .013402
_cons	.2168544	.108595	2.00	0.046	.0035183 .4301904

```
. reg LNWAGE EDU EXPER
```

Source	SS	df	MS	Number of obs	=	526
Model	36.9850401	2	18.4925201	F(2, 523)	=	86.86
Residual	111.34471	523	.212896195	Prob > F	=	0.0000
				R-squared	=	0.2493
				Adj R-squared	=	0.2465
Total	148.32975	525	.282532857	Root MSE	=	.46141

LNWAGE	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]
EDUC	.0979356	.0076224	12.85	0.000	.0829613 .1129099
EXPER	.0103469	.0015551	6.65	0.000	.0072919 .013402
_cons	.2168544	.108595	2.00	0.046	.0035183 .4301904

拒绝原假设，工资对教育的半弹性比对经验的半弹性系数高出8.7%，且具有统计显著性



多元线性回归模型的 设定、误设和诊断

模型设定问题

- 如何设置多元模型
 - 基于理论成果和实际问题确定自变量
 - 基于理论成果，注意自变量自动筛选策略的应用场合
 - 如何控制混淆变量并没有明确的说明，存在主观偏差和模型依赖
 - 基于实际问题设置交互项
- 模型误设问题的三种情况
 - 模型过度设定
 - 模型设定不足
 - 函数形式误设

多元模型的交互项设置问题

- 多元线性回归模型默认两个基本假设：

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \dots + \beta_k x_k + u.$$

- 线性 (Linear)

$$E(y | \mathbf{x}) = \beta_0 + \beta_1 g_1(\mathbf{x}) + \beta_2 g_2(\mathbf{x}) + \dots + \beta_M g_M(\mathbf{x})$$

$$\text{we define } z_1 \equiv g_1(\mathbf{x}), \dots, z_M \equiv g_M(\mathbf{x}),$$

$$E(y | z_1, z_2, \dots, z_M) = \beta_0 + \beta_1 z_1 + \beta_2 z_2 + \dots + \beta_M z_M$$

- 可加性 (Additive)

- x_i 对 y 效应具有可加性： x_i 对 y 效应的变化独立于 x_j 对 y 的

■ 例：从对商品销售量的影响效应角度，研究不同媒体(TV, Radio)的广告投放策略

	TV	Radio	Newspaper	Sales
1	230.1	37.8	69.2	22.1
2	44.5	39.3	45.1	10.4
3	17.2	45.9	69.3	9.3
4	151.5	41.3	58.5	18.5
5	180.8	10.8	58.4	12.9
6	8.7	48.9	75	7.2
7	57.5	32.8	23.5	11.8
8	120.2	19.6	11.6	13.2
9	8.6	2.1	1	4.8
10	199.8	2.6	21.2	10.6
11	66.1	5.8	24.2	8.6

. reg Sales TV Radio

Source	SS	df	MS	Number of obs	=	200
Model	4860.23477	2	2430.11738	F(2, 197)	=	859.62
Residual	556.91398	197	2.82697452	Prob > F	=	0.0000
Total	5417.14875	199	27.221853	R-squared	=	0.8972
				Adj R-squared	=	0.8962
				Root MSE	=	1.6814

Sales	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]
TV	.0457548	.0013904	32.91	0.000	.0430129 .0484967
Radio	.1879942	.00804	23.38	0.000	.1721388 .2038497
_cons	2.9211	.2944897	9.92	0.000	

> summary(Adv_lm)

```
Call:
lm(formula = Sales ~ TV + Radio, data = Data)

Residuals:
    Min       1Q   Median       3Q      Max
-8.7977 -0.8752  0.2422  1.1708  2.8328

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  2.92110    0.29449   9.919  <2e-16 ***
TV           0.04575    0.00139  32.909  <2e-16 ***
Radio        0.18799    0.00804  23.382  <2e-16 ***
---

```

> library(lm.beta)

> lm.beta(Adv_lm)

Call:

lm(formula = Sales ~ TV + Radio, data = Data)

Standardized Coefficients::

```
(Intercept)      TV      Radio
0.0000000      0.7529042      0.5349569
```

- TV和Radio对Sales有显著影响，但TV的效应小于Radio效应，对吗？

- 计算标准化回归系数

. reg Sales TV Radio, beta

Source	SS	df	MS	Number of obs	=	200
Model	4860.23477	2	2430.11738	F(2, 197)	=	859.62
Residual	556.91398	197	2.82697452	Prob > F	=	0.0000
Total	5417.14875	199	27.221853	R-squared	=	0.8972
				Adj R-squared	=	0.8962
				Root MSE	=	1.6814

Sales	Coef.	Std. Err.	t	P> t	Beta
TV	.0457548	.0013904	32.91	0.000	.7529042
Radio	.1879942	.00804	23.38	0.000	.5349569
_cons	2.9211	.2944897	9.92	0.000	.

- 是否减少甚至停止Radio投放？
- Radio一般会促进TV对Sales的效应(不独立)
 - 经济学认为：一定投入预算下，多方面的投入将比单方面的投入带来更高的回报，具有协同效应(synergy effect)
 - 协同效应可通过交互项(交互效应)刻画

■ 考虑协同效应时建立带交互项的多元回归模型

$$\begin{aligned}\text{sales} &= \beta_0 + \beta_1 \times \text{TV} + \beta_2 \times \text{radio} + \beta_3 \times (\text{radio} \times \text{TV}) + \epsilon \\ &= \beta_0 + (\beta_1 + \beta_3 \times \text{radio}) \times \text{TV} + \beta_2 \times \text{radio} + \epsilon.\end{aligned}$$

- 不能停止radio投放的原因：radio投放增加一个单位使sales增加 β_2 的同时，也使TV投放对sales的效应增加 β_3 ，等

. reg Sales TV Radio c.TV#c.Radio, beta

Source	SS	df	MS	Number of obs	=	200
Model	5242.66537	3	1747.55512	F(3, 196)	=	1963.06
Residual	174.483383	196	.89022134	Prob > F	=	0.0000
Total	5417.14875	199	27.221853	R-squared	=	0.9678
				Adj R-squared	=	0.9673
				Root MSE	=	.94352

Sales	Coef.	Std. Err.	t	P> t	Beta
TV	.0191011	.0015041	12.70	0.000	.3143118
Radio	.0288603	.0089053	3.24	0.001	.0821251
c.TV#c.Radio	.0010865	.0000524	20.73	0.000	.699848
_cons	6.75022	.2478714	27.23	0.000	

较强的协同效应

lm(formula = Sales ~ TV + Radio + TV:Radio, data = Data)

Residuals:

Min	1Q	Median	3Q	Max
-6.3366	-0.4028	0.1831	0.5948	1.5246

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	6.750e+00	2.479e-01	27.233	<2e-16 ***
TV	1.910e-02	1.504e-03	12.699	<2e-16 ***
Radio	2.886e-02	8.905e-03	3.241	0.0014 **
TV:Radio	1.086e-03	5.242e-05	20.727	<2e-16 ***

Standardized Coefficients::

(Intercept)	TV	Radio	TV:Radio
0.00000000	0.31431179	0.08212506	0.69984796

通常，若交互效应(高阶效应)显著但主效应(Radio, 低阶效应)不显著时，仍倾向保留主效应

多元模型的交互项设定问题

- 再看带交互项的多元回归模型

- 例：研究就职年数对年收入的因果影响, 需控制职位, 且年数对收入的影响还取决于职位(存在交互效应)

$$inc = \beta_0 + \beta_1 years + \beta_2 post + \beta_3 years.post + u$$

- 自变量 (years) 的偏效应近似为：

$$\frac{\Delta inc}{\Delta years} \approx \beta_1 + \beta_3 post$$

- 自变量的偏效应不仅取决于 β_1 ，还取决于交互项参数(β_3 大于零表示职位越高，多就职一年收入增加越多)
- years的偏效应还取决于post的取值
 - 一般可代入**均值**
- **问题：** years偏效应的显著性如何？

- 为便于解释偏效应，含交互项的一般建模策略
 - 便于检验显著性：引入两变量减各自均值后的交互项

$$inc = \beta_0 + \beta_1 years + \beta_2 post + \beta_3 years.post + u$$

$$\frac{\Delta inc}{\Delta years} \approx \beta_1 + \beta_3 post$$

$$\delta_1 = \beta_1 + \beta_3 \mu_2, \delta_2 = \beta_2 + \beta_3 \mu_1$$

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1 x_2 + u$$



调整为(第4章)

$$\delta_1 = \beta_1 + \beta_3 \mu_2, \delta_2 = \beta_2 + \beta_3 \mu_1$$

$$\beta_1 = \delta_1 - \beta_3 \mu_2, \beta_2 = \delta_2 - \beta_3 \mu_1$$

$$y = \beta_0 + (\delta_1 - \beta_3 \mu_2)x_1 + (\delta_2 - \beta_3 \mu_1)x_2 + \beta_3 x_1 x_2 + u$$

$$y = \beta_0 + \delta_1 x_1 + \delta_2 x_2 + \beta_3 (x_1 x_2 - x_1 \mu_2 - x_2 \mu_1) + u$$

$$y = \beta_0 - \beta_3 \mu_1 \mu_2 + \delta_1 x_1 + \delta_2 x_2 + \beta_3 (x_1 - \mu_1)(x_2 - \mu_2) + u$$

$$y = \alpha_0 + \delta_1 x_1 + \delta_2 x_2 + \beta_3 (x_1 - \mu_1)(x_2 - \mu_2) + u$$

- 例：从对商品销售量的影响效应角度，研究不同媒体(TV, Radio)的广告投放策略

$$y = \alpha_0 + \delta_1 x_1 + \delta_2 x_2 + \beta_3 (x_1 - \mu_1)(x_2 - \mu_2) + \mu$$

```
> TV_Radio<-with(Data, (TV-mean(TV)) * (Radio-mean(Radio)))  
> Adv_lm<-lm(Sales~TV+Radio+TV_Radio,data=Data)  
> summary(Adv_lm)
```

```
Call:  
lm(formula = Sales ~ TV + Radio + TV_Radio, data = Data)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-6.3366	-0.4028	0.1831	0.5948	1.5246

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	3.034e+00	1.653e-01	18.35	<2e-16	***
TV	4.438e-02	7.830e-04	56.67	<2e-16	***
Radio	1.886e-01	4.512e-03	41.81	<2e-16	***
TV_Radio	1.086e-03	5.242e-05	20.73	<2e-16	***

```
lm(formula = Sales ~ TV + Radio + TV:Radio, data = Data)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-6.3366	-0.4028	0.1831	0.5948	1.5246

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	6.750e+00	2.479e-01	27.233	<2e-16	***
TV	1.910e-02	1.504e-03	12.699	<2e-16	***
Radio	2.886e-02	8.905e-03	3.241	0.0014	**
TV:Radio	1.086e-03	5.242e-05	20.727	<2e-16	***

- 例：从对商品销售量的影响效应角度，研究不同媒体(TV, Radio)的广告投放策略

$$y = \alpha_0 + \delta_1 x_1 + \delta_2 x_2 + \beta_3 (x_1 - \mu_1)(x_2 - \mu_2) + \mu$$

```
. egen meanTV=mean(TV)
```

```
. egen meanRadio=mean(Radio)
```

```
. gen TV_Radio=(TV-meanTV)*(Radio-meanRadio)
```

```
. reg Sales TV Radio TV_Radio
```

Source	SS	df	MS	Number of obs	=	200
Model	5242.66537	3	1747.55512	F(3, 196)	=	1963.06
Residual	174.483381	196	.890221333	Prob > F	=	0.0000
				R-squared	=	0.9678
				Adj R-squared	=	0.9673
Total	5417.14875	199	27.221853	Root MSE	=	.94352

Sales	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
TV	.0443773	.000783	56.67	0.000	.042833	.0459216
Radio	.1886212	.0045118	41.81	0.000	.1797233	.1975192
TV_Radio	.0010865	.0000524	20.73	0.000	.0009831	.0011899
_cons	3.033543	.1653454	18.35	0.000	2.707458	3.359627

```
. reg Sales TV Radio c.TV#c.Radio, beta
```

Source	SS	df	MS	Number of obs	=	200
Model	5242.66537	3	1747.55512	F(3, 196)	=	1963.06
Residual	174.483383	196	.89022134	Prob > F	=	0.0000
				R-squared	=	0.9678
				Adj R-squared	=	0.9673
Total	5417.14875	199	27.221853	Root MSE	=	.94352

Sales	Coef.	Std. Err.	t	P> t	Beta
TV	.0191011	.0015041	12.70	0.000	.3143118
Radio	.0288603	.0089053	3.24	0.001	.0821251
c.TV#c.Radio	.0010865	.0000524	20.73	0.000	.699848
_cons	6.75022	.2478714	27.23	0.000	.

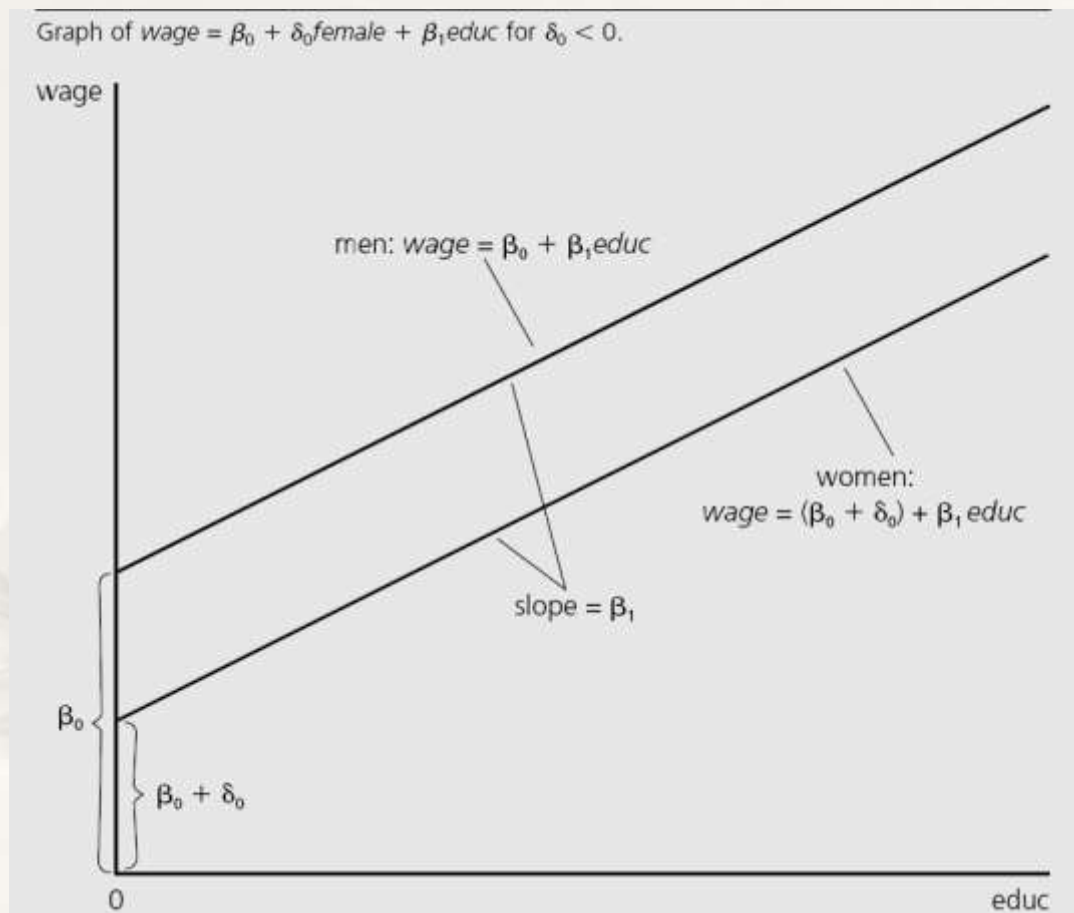
- 带交互项的多元回归模型中，交互变量也可以是分类变量
 - 虚拟自变量的回归

$$wage = \beta_0 + \delta_0 female + \beta_1 educ + u.$$

<i>person</i>	<i>wage</i>	<i>educ</i>	<i>exper</i>	<i>female</i>	<i>married</i>
1	3.10	11	2	1	0
2	3.24	12	22	1	1
3	3.00	11	2	0	0
4	6.00	8	44	0	1
5	5.30	12	7	0	1
.	.	.	.	.	.
.	.	.	.	.	.
.	.	.	.	.	.
525	11.56	16	5	0	1
526	3.50	14	5	1	0

- 虚拟变量是一个二值(binary)变量
 - 用于表述定性变量
- 这里的Female是一个性别的虚拟变量
 - 虚拟变量female，取1表示女，取0表示男(0为基准组或参照类)，将与基准组(参照类)对比
- 若有 k 个分组水平，只能设置 $k - 1$ 个虚拟变量，否则产生自变量完全共线性问题，称虚拟变量陷阱(dummy variable trap)

■ 虚拟自变量回归示例：考察受教育水平相同条件下，性别对工资收入的影响(第7章)



$$wage = \beta_0 + \delta_0 female + \beta_1 educ + u.$$

$$\delta_0 = E(wage|female, educ) - E(wage|male, educ).$$

- δ_0 是在相同受教育程度下性别差异所致的平均工资收入的差异
- 两方程有相同的斜率，不同性别组中受教育程度带来的工资变化是相同的

- 例：考察不同性别的工资情况和教育回报率
 - 男性为基准组：

```
contrasts(Data$FEMALE) #男性为参照类
lgWAGE_lm2<-lm(log(WAGE)~EDUC+FEMALE,data=Data)
summary(lgWAGE_lm2)
Data$FEMALE<-relevel(Data$FEMALE,ref="1") #指定女性为参照类
contrasts(Data$FEMALE) #女性为参照类
lgWAGE_lm2<-lm(log(WAGE)~EDUC+FEMALE,data=Data)
summary(lgWAGE_lm2)
```

```
> lgWAGE_lm2<-lm(log(WAGE)~EDUC+FEMALE,data=Data)
> summary(lgWAGE_lm2)

Call:
lm(formula = log(WAGE) ~ EDUC + FEMALE, data = Data)
```

Residuals:

Min	1Q	Median	3Q	Max
-2.02672	-0.27470	-0.03731	0.26219	1.34738

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.826269	0.094054	8.785	<2e-16 ***
EDUC	0.077203	0.007047	10.955	<2e-16 ***
FEMALE1	-0.360865	0.039024	-9.247	<2e-16 ***

```
Data<-read.table(file="wage1.txt",header=TRUE,sep=" ")
###WAGE:average hourly earnings; EDUC:years of education
#EXPER:years potential experience; TENURE:years with current employer
#SMSA:standard metropolitan statistical area
#FEMALE: =1 if female; NONWHITE: =1 if nonwhite
```

```
Data$FEMALE<-as.factor(Data$FEMALE)
```

R将默认因子的最低水平为基准组

女性为基准时：

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.465404	0.091227	5.102	4.72e-07 ***
EDUC	0.077203	0.007047	10.955	< 2e-16 ***
FEMALE0	0.360865	0.039024	9.247	< 2e-16 ***

. reg LNWAGE EDU FEMALE

Source	SS	df	MS	Number of obs	=	526
Model	44.5315183	2	22.2657591	F(2, 523)	=	112.19
Residual	103.798232	523	.198466982	Prob > F	=	0.0000
Total	148.32975	525	.282532857	R-squared	=	0.3002
				Adj R-squared	=	0.2975
				Root MSE	=	.4455

LNWAGE	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]
EDUC	.0772033	.0070472	10.96	0.000	.0633591 .0910475
FEMALE	-.3608654	.0390245	-9.25	0.000	-.4375294 -.2842015
_cons	.8262693	.0940541	8.79	0.000	.6414991 1.01104

- 例：考察不同性别的工资情况和教育回报
 - 通常男性和女性的教育回报率可能不相同，即两条回归直线的斜率可能不同：男女分别建模

```
> lgWAGE_sep<-lm(log(WAGE)~FEMALE/EDUC-1,data=Data)
> summary(lgWAGE_sep)
```

```
Call:
lm(formula = log(WAGE) ~ FEMALE/EDUC - 1, data = Data)
```

```
Residuals:
    Min       1Q   Median       3Q      Max
-2.02673 -0.27468 -0.03721  0.26221  1.34740
```

```
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
FEMALE0      0.825955   0.118050   6.997 8.08e-12 ***
FEMALE1      0.465890   0.142997   3.258 0.00119 **
FEMALE0:EDUC  0.077228   0.008988   8.593 < 2e-16 ***
FEMALE1:EDUC  0.077164   0.011383   6.779 3.29e-11 ***
```

```
> cofMatrix
              (Intercept)          EDUC
MALE      0.8259546 0.07722794
FEMALE    0.4658902 0.07716385
```

- 问：男女的教育回报率差异具有统计显著性吗？

by FEMALE, sort : regress LNWAGE EDUC

> FEMALE = 0

Source	SS	df	MS	Number of obs	=	274
Model	14.682095	1	14.682095	F(1, 272)	=	62.99
Residual	63.4020485	272	.233095767	Prob > F	=	0.0000
				R-squared	=	0.1880
				Adj R-squared	=	0.1850
Total	78.0841435	273	.286022504	Root MSE	=	.4828

LNWAGE	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]
EDUC	.0772279	.0097308	7.94	0.000	.0580707 .0963852
_cons	.8259546	.1278128	6.46	0.000	.5743265 1.077583

> FEMALE = 1

Source	SS	df	MS	Number of obs	=	252
Model	9.13742695	1	9.13742695	F(1, 250)	=	56.55
Residual	40.3961794	250	.161584718	Prob > F	=	0.0000
				R-squared	=	0.1845
				Adj R-squared	=	0.1812
Total	49.5336064	251	.197345045	Root MSE	=	.40198

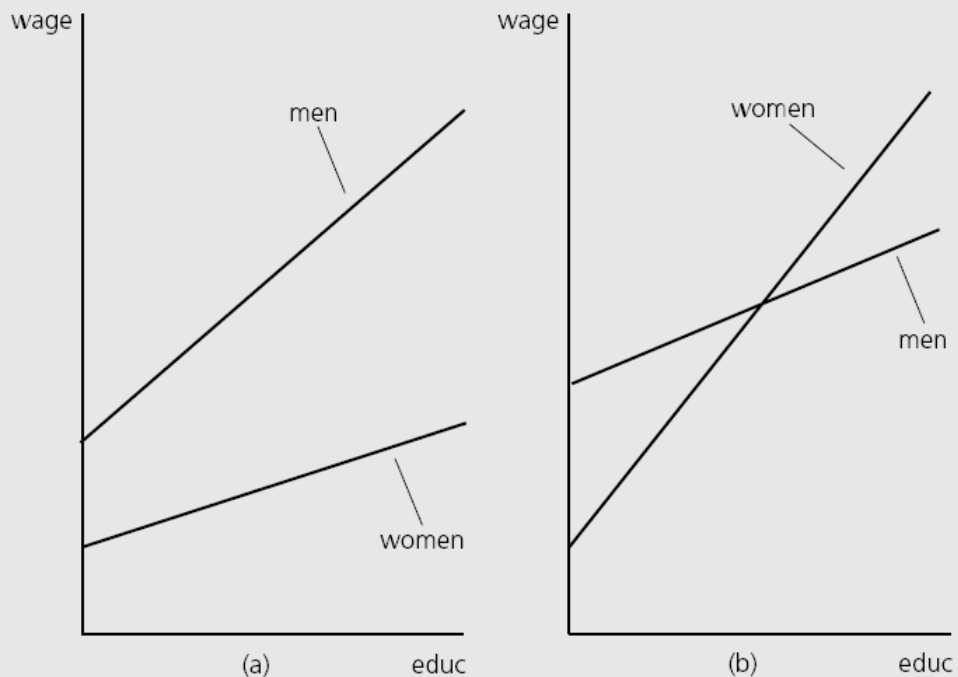
LNWAGE	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]
EDUC	.0771639	.0102613	7.52	0.000	.0569543 .0973735
_cons	.4658902	.1289047	3.61	0.000	.2120126 .7197677

- 例：考察不同性别的工资情况和教育回报
 - 引入交互项(第7章)

$$\log(wage) = \beta_0 + \delta_0 female + \beta_1 educ + \delta_1 female \cdot educ + u.$$

$$\log(wage) = (\beta_0 + \delta_0 female) + (\beta_1 + \delta_1 female)educ + u.$$

Graphs of equation (7.16). (a) $\delta_0 < 0, \delta_1 < 0$; (b) $\delta_0 < 0, \delta_1 > 0$.



- δ_0 度量了性别在工资变化百分比上的差距;
- δ_1 度量了不同性别在教育回报率上的差异;

```
. reg LNWAGE EDU FEMALE c.EDU#c.FEMALE
```

Source	SS	df	MS	Number of obs	=	526
Model	44.5315221	3	14.8438407	F(3, 522)	=	74.65
Residual	103.798228	522	.19884718	Prob > F	=	0.0000
Total	148.32975	525	.282532857	R-squared	=	0.3002
				Adj R-squared	=	0.2962
				Root MSE	=	.44592

LNWAGE	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
EDUC	.0772279	.0089875	8.59	0.000	.0595718	.0948841
FEMALE	-.3600645	.1854296	-1.94	0.053	-.7243444	.0042154
c. EDUC#c. FEMALE	-.0000641	.0145035	-0.00	0.996	-.0285565	.0284283
_cons	.8259546	.1180502	7.00	0.000	.5940427	1.057867

交互效应
不显著

- 例：考察不同性别的工资情况和教育回报
 - 引入交互项(第7章)

```
lm(formula = log(WAGE) ~ EDUC + FEMALE + EDUC:FEMALE, data = Data)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-2.02673	-0.27468	-0.03721	0.26221	1.34740

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	8.260e-01	1.181e-01	6.997	8.08e-12 ***
EDUC	7.723e-02	8.988e-03	8.593	< 2e-16 ***
FEMALE1	-3.601e-01	1.854e-01	-1.942	0.0527 .
EDUC:FEMALE1	-6.408e-05	1.450e-02	-0.004	<u>0.9965</u>

```
lgWAGE_lm3<-lm(log(WAGE)~EDUC+FEMALE+EDUC:FEMALE,data=Data)
```

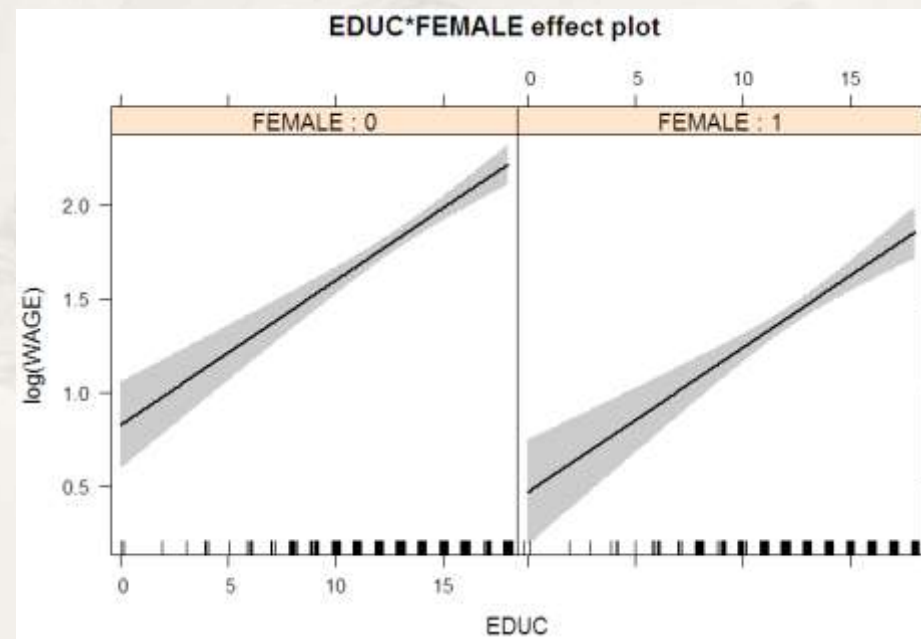
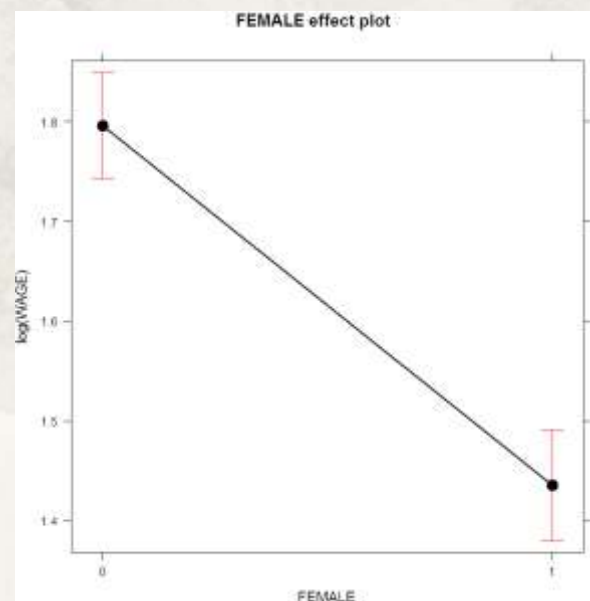
```
summary(lgWAGE_lm3) #交互效应不显著
```

```
library("effects")
```

```
plot(effect("FEMALE",lgWAGE_lm3))
```

```
plot(allEffects(lgWAGE_lm3,default.levels=50)) #因变量的置信区间
```

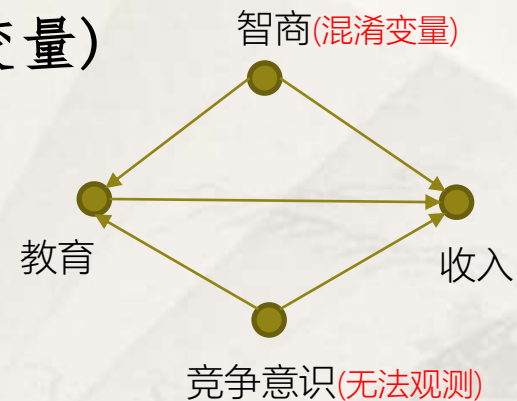
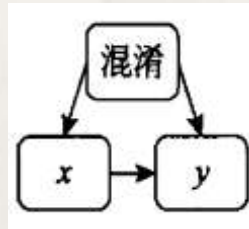
交互效应不显著



交互项与多元回归中的变量角色

- **混淆变量**: 是影响自变量和因变量的因素，会干扰研究者分析自变量对因变量的影响，是需施加**控制**或排除的变量

- 例：年轻人长相对结婚年龄的影响(地域为混淆变量)



- 混淆变量通常是自变量的**前置变量**

- 例：研究学历对收入的影响时职业作为混淆变量？学历会影响一个人的职业，即职业并不一定是学历的前置变量(并非是混淆变量，但均时需加以控制的变量)

- 自变量以外的其他影响因变量的变量都需被控制，除混淆变量外典型的还包括：

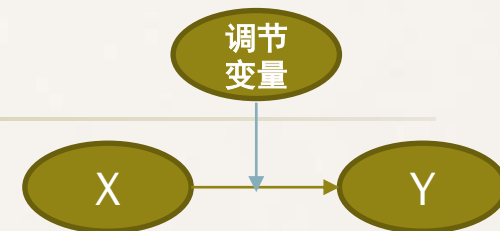
- **调节变量；中介变量**

- **调节变量**: 若变量 y 与 x 的关系是变量 M 的函数, 则称 M 为调节变量

- 调节变量通过**交互项**的形式表述

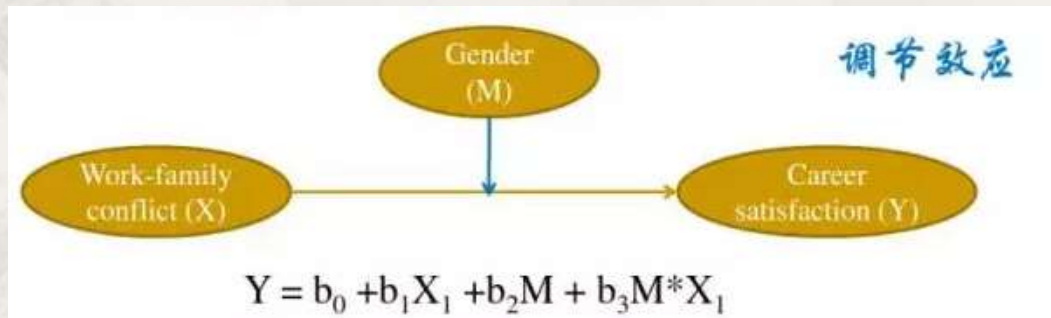
- 例:

$$\begin{aligned}\text{sales} &= \beta_0 + \beta_1 \times \text{TV} + \beta_2 \times \text{radio} + \beta_3 \times (\text{radio} \times \text{TV}) + \epsilon \\ &= \beta_0 + (\beta_1 + \beta_3 \times \text{radio}) \times \text{TV} + \beta_2 \times \text{radio} + \epsilon.\end{aligned}$$



- 再例:

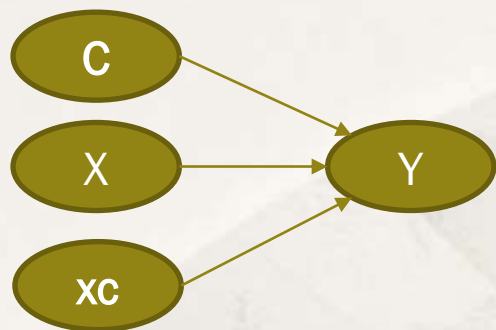
radio 可视为调节变量, β_3 代表调节效应



b_3 代表调节效应

- 调节变量影响因变量和自变量间关系的方向和强弱
 - **增强型**调节效应和**干扰型**调节效应
 - 有时主效应不显著但交互项显著, 为关注调节效应可保留主效应

- 调节变量：调节变量通过交互项的形式表述，但逻辑上与交互项并不完全相同
 - 交互项分析中：两变量的地位可以是对等的，其中任一个都可作为调节变量；交互项并不区分哪个是调节变量



$$\begin{aligned} \text{sales} &= \beta_0 + \beta_1 \times \text{TV} + \beta_2 \times \text{radio} + \beta_3 \times (\text{radio} \times \text{TV}) + \epsilon \\ &= \beta_0 + (\beta_1 + \beta_3 \times \text{radio}) \times \text{TV} + \beta_2 \times \text{radio} + \epsilon. \end{aligned}$$

radio 可视作为调节变量

$$\text{sales} = \beta_0 + \beta_1 \text{TV} + (\beta_2 + \beta_3 \text{TV}) \times \text{radio} + \epsilon \quad \text{TV 可视作为调节变量}$$

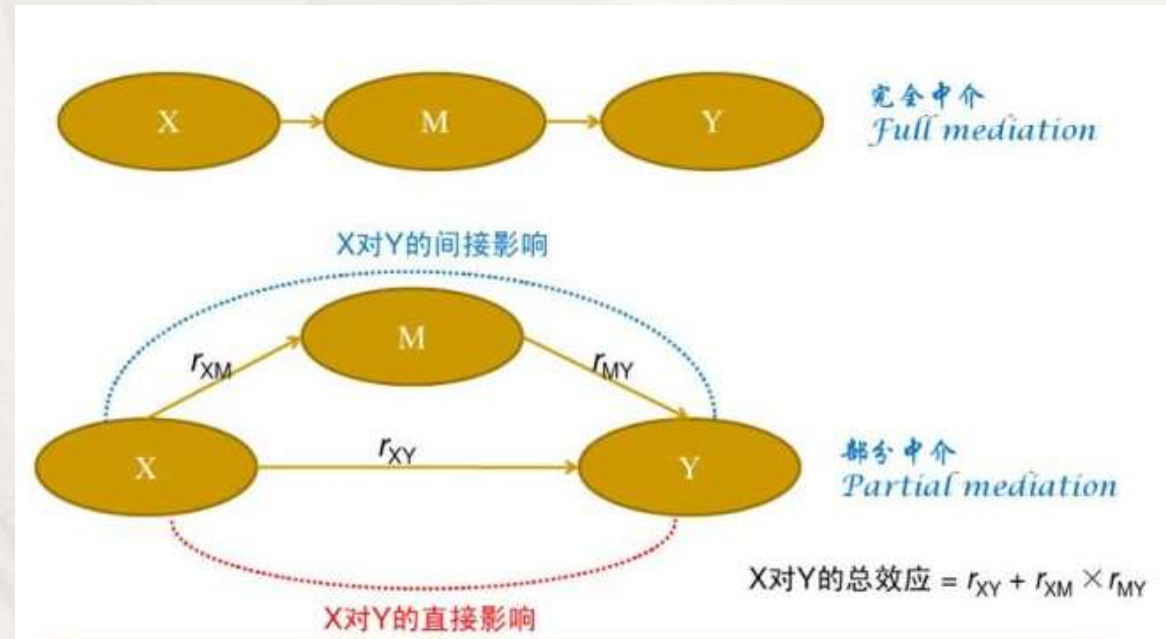
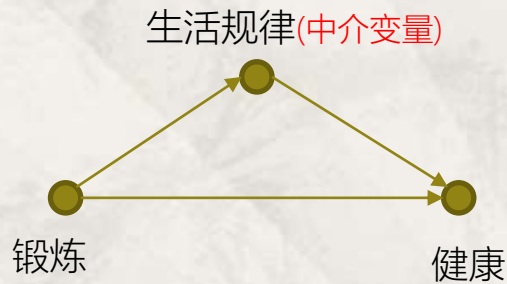
- 调节效应分析中：哪个是自变量，哪个是调节变量是明确的，在一个确定的研究中两者不能互换
 - 例：研究性别在数学能力上的差异时，基于每个年级各用一份测试题的得分数据研究——**年级作为调节变量**

$$\text{scores} = \beta_0 + \beta_1 \text{gender} + \beta_2 \text{Grade} + \beta_3 \text{gender} \times \text{Grade} + \epsilon \quad \longrightarrow \quad \text{scores} = \beta_0 + (\beta_1 + \beta_3 \text{Grade}) \times \text{gender} + \beta_2 \text{Grade} + \epsilon$$

- 该数据不支持：研究年级在数学能力上的差异(性别作为调节变量)，因为不同年级的测试题不同，得分不具可比性

交互项与多元回归中的变量角色

- **中介变量**：如果 x 通过影响变量 M 来影响 y ，则称 M 为中介变量



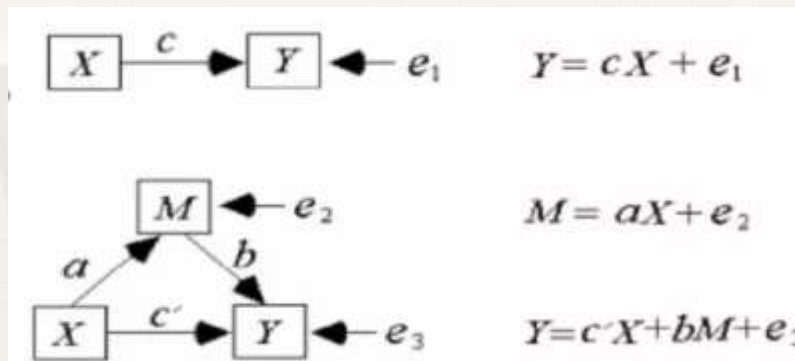
- 若一个变量，与自变量或因变量不相关，不可能成为中介变量，但有可能成为调节变量(例前例中的年级，作为调节变量与性别不相关)

- 中介变量的作用：研究 x 对 y 影响的总效应和影响机制
 - 例，“家庭经济地位(x)”对“家庭功能(M)”的影响进而影响“青少年疏离感(y)”，直接效应是多少？中介效应是多少？

■ 中介模型(数据中心化):

研究目标：
估计总效应 c

$$\begin{aligned} Y &= cX + e_1 \\ M &= aX + e_2 \\ Y &= c'X + bM + e_3 \end{aligned}$$



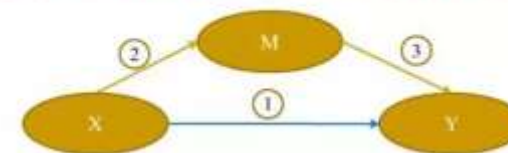
- $Y = c'X + b(aX + e_2) + e_3 = (c' + ab)X + (be_2 + e_3)$

中介变量 M 并不以交互项形式出现

- 称： c 为总效应($c = c' + ab$), c' 为直接效应, ab 为中介效应

- 第一步：估计和检验总效应(c)
- 第二步：通过两小步估计和检验中介效应(ab)
 - 可采用bootstrap得到若干个 ab 的估计值，并得到95%的置信区间(R包mediation的mediate函数实现)
- 第三步：基于 c' 检验完全中介还是部分中介

使用回归方法检验中介作用



完全中介:

① $Y = b_0 + b_1 * X$

② $M = b_2 + b_3 * X$

③ $Y = b_4 + b_5 X + b_6 * M$

部分中介:

① $Y = b_0 + b_1 * X$

② $M = b_2 + b_3 * X$

③ $Y = b_4 + b_5 * X + b_6 * M$

Note: * 显著

模型设定问题

- 如何设置多元模型
 - 基于理论成果和实际问题确定变量
 - 基于理论成果，注意自变量自动筛选策略的应用场合
 - 如何控制混淆变量并没有明确的说明，存在主观偏差和模型依赖
 - 基于实际问题设置交互项
- 模型误设问题的三种情况
 - 模型过度设定
 - 模型设定不足
 - 函数形式误设

模型误设问题

- **模型过度设定**: 指模型包含了 **需被控制** 的对 y 没有显著线性 **净** 影响的变量 (3.4节)

- 表现: 回归系数显著性检验无法拒绝原假设: $H_0: \beta_j = 0$
- 通常策略: 剔除不显著的变量, 得到简单模型

- 选择简单模型对自变量效应估计的影响

- 情况一: 与自变量 **无关** (如前例中的年级, 调节变量)

$$scores = \beta_0 + (\beta_1 + \beta_3 \text{Grade}) \times \text{gender} + \beta_2 \text{Grade} + \varepsilon$$

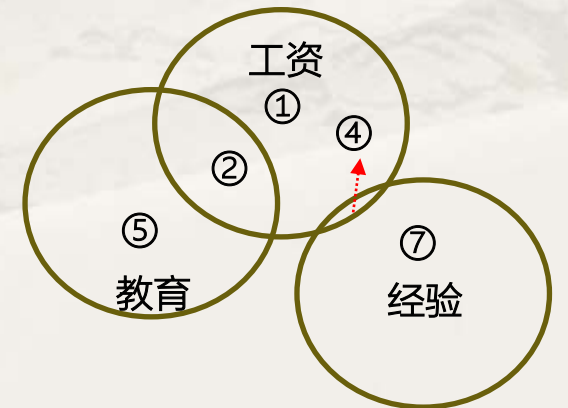
$$wage = \beta_0 + \beta_1 \text{edu} + \beta_2 \text{exper} + u$$

$$wage = \beta_0 + \beta_1 \text{edu} + u$$

- 满足假定1-3条件下, 无论变量对 y 是否有无影响, 对自变量的效应估计都满足无偏性, 即:

$$E(\hat{\beta}_j) = \beta_j, j = 0, 1, \dots, k,$$

- 选择简单模型, 且可获得无偏估计

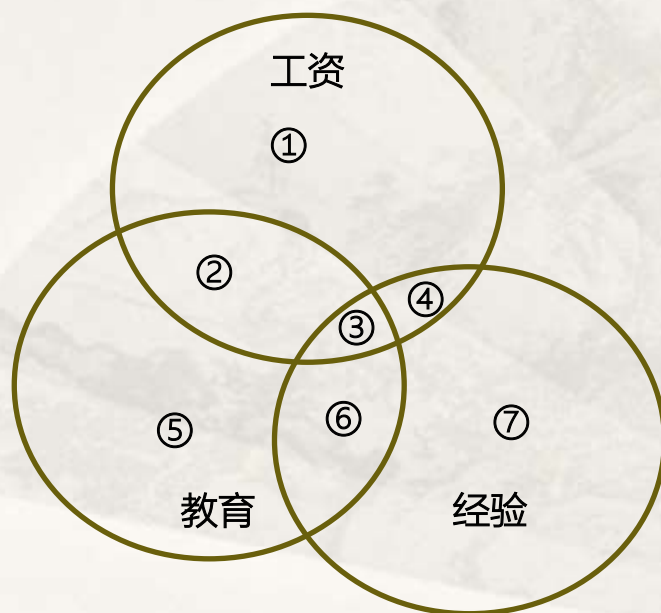


■ 选择简单模型对效应估计的影响：

■ 情况二：与自解释变量相关

$$wage = \beta_0 + \beta_1 edu + \beta_2 exper + u$$

$$wage = \beta_0 + \beta_1 edu + u$$



■ 帽为过度设定模型 (无偏估计); 弯为简单模型 (存在变量遗漏, 有偏估计)

$$\text{Var}(\hat{\beta}_1) = \sigma^2 / [\text{SST}_1 (1 - R_1^2)], \quad \text{Var}(\tilde{\beta}_1) = \sigma^2 / \text{SST}_1$$

↓

$$\text{Var}(\tilde{\beta}_1) < \text{Var}(\hat{\beta}_1).$$

■ 过度设置中的多重共线性, 会使估计量的方差增大。但不引入则估计有偏

■ 无偏大方差和有偏小方差的选择问题

■ 关注: 偏差③的大小

■ **偏差**是多少？多元模型(过度设置模型:无偏估计)与一元模型(简单模型:有偏估计)的估计:

■ 一元回归: $\tilde{y} = \tilde{\beta}_0 + \tilde{\beta}_1 x_1$

■ 多元回归: $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2$ 两个不同的估计值

```
> compareCoefs(lgWAGE_lm1,lgWAGE_lm2) #直观对比两个模型的系数
```

Call:

```
1:"lm(formula = log(WAGE) ~ EDUC, data = Data)"
```

```
2:"lm(formula = log(WAGE) ~ EDUC + EXPER, data = Data)"
```

	Est. 1	SE 1	Est. 2	SE 2
(Intercept)	0.58377	0.09734	0.21685	0.10860
EDUC	0.08274	0.00757	0.09794	0.00762
EXPER			0.01035	0.00156

$$wage = \beta_0 + \beta_1 edu + u$$

$$wage = \beta_0 + \beta_1 edu + \beta_2 exper + u$$

```
exper_lm<-lm(EXPER~EDUC,data=Data)
```

$$\tilde{\beta}_1 = \hat{\beta}_1 + \hat{\beta}_2 \tilde{\delta}_1$$

$$0.08274 = 0.09794 + 0.01035 (-1.4682)$$

```
> summary(exper_lm)
```

Call:

```
lm(formula = EXPER ~ EDUC, data = Data)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-20.652	-9.971	-2.971	9.125	30.625

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	35.4615	2.6279	13.494	< 2e-16 ***
EDUC	-1.4682	0.2043	-7.187	2.3e-12 ***

■ 偏差是多少？多元模型 (过度设置模型: 无偏估计) 与一元模型 (简单模型: 有偏估计) 的估计:

■ 一元回归: $\tilde{y} = \tilde{\beta}_0 + \tilde{\beta}_1 x_1$

■ 多元回归: $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2$

$$wage = \beta_0 + \beta_1 edu + u$$

两个不同的估计值

$$wage = \beta_0 + \beta_1 edu + \beta_2 exper + u$$

. reg EXPER EDU

Source	SS	df	MS	Number of obs	=	526
Model	8677.05239	1	8677.05239	F(1, 524)	=	51.65
Residual	88029.7936	524	167.995789	Prob > F	=	0.0000
				R-squared	=	0.0897
				Adj R-squared	=	0.0880
Total	96706.846	525	184.203516	Root MSE	=	12.961

EXPER	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]
EDUC	-1.468182	.2042881	-7.19	0.000	-1.869507 -1.066858
_cons	35.4615	2.627905	13.49	0.000	30.29898 40.62402

$$\tilde{\beta}_1 = \hat{\beta}_1 + \hat{\beta}_2 \tilde{\delta}_1$$

$$0.08274 = 0.09794 + 0.01035 (-1.4682)$$

. reg LNWAGE EDU

Source	SS	df	MS	Number of obs	=	526
Model	27.5606291	1	27.5606291	F(1, 524)	=	119.58
Residual	120.769121	524	.230475422	Prob > F	=	0.0000
				R-squared	=	0.1858
				Adj R-squared	=	0.1843
Total	148.32975	525	.282532857	Root MSE	=	.48008

LNWAGE	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]
EDUC	.0827444	.0075667	10.94	0.000	.0678796 .0976092
_cons	.5837727	.0973358	6.00	0.000	.3925563 .7749891

. reg LNWAGE EDU EXPER

Source	SS	df	MS	Number of obs	=	526
Model	36.9850401	2	18.4925201	F(2, 523)	=	86.86
Residual	111.34471	523	.212896195	Prob > F	=	0.0000
				R-squared	=	0.2493
				Adj R-squared	=	0.2465
Total	148.32975	525	.282532857	Root MSE	=	.46141

LNWAGE	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]
EDUC	.0979356	.0076224	12.85	0.000	.0829613 .1129099
EXPER	.0103469	.0015551	6.65	0.000	.0072919 .013402
_cons	.2168544	.108595	2.00	0.046	.0035183 .4301904

■ **偏差**是多少？多元模型 (过度设置模型: 无偏估计) 与一元模型 (简单模型: 有偏估计) 的估计:

■ 一元回归:

$$\tilde{y} = \tilde{\beta}_0 + \tilde{\beta}_1 x_1$$

■ 多元回归:

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2$$

■ 两个估计的关系:

$$\tilde{\beta}_1 = \hat{\beta}_1 + \hat{\beta}_2 \tilde{\delta}_1 \quad \delta_1 \text{ 是舍弃的变量 } x_2 \text{ 对自变量 } x_1 \text{ 的回归系数}$$

依迭代期望法则:

$$E(y | \mathbf{x}) = E[E(y | \mathbf{x}, \mathbf{z}) | \mathbf{x}]$$

$$\text{or, defining } \mu_1(\mathbf{x}, \mathbf{z}) \equiv E(y | \mathbf{x}, \mathbf{z}) \text{ and } \mu_2(\mathbf{x}) \equiv E(y | \mathbf{x}),$$

回归模型中, 例:

$$\mu_2(\mathbf{x}) = E[\mu_1(\mathbf{x}, \mathbf{z}) | \mathbf{x}]$$

$$E(y | x_1, x_2, z) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 z$$

无偏估计

$$E(z | x_1, x_2) = \delta_0 + \delta_1 x_1 + \delta_2 x_2$$

$$E(y | x_1, x_2) = E(\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 z | x_1, x_2)$$

$$= \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 E(z | x_1, x_2)$$

$$= \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 (\delta_0 + \delta_1 x_1 + \delta_2 x_2)$$

$$= (\beta_0 + \beta_3 \delta_0) + (\beta_1 + \beta_3 \delta_1) x_1 + (\beta_2 + \beta_3 \delta_2) x_2$$

有偏估计 $\tilde{\beta}_1$

δ_1 是舍弃的变量 z 对自变量 x_1 和变量 x_2 的回归系数

$$wage = \beta_0 + \beta_1 edu + u \quad wage = \beta_0 + \beta_1 edu + \beta_2 exper + u$$

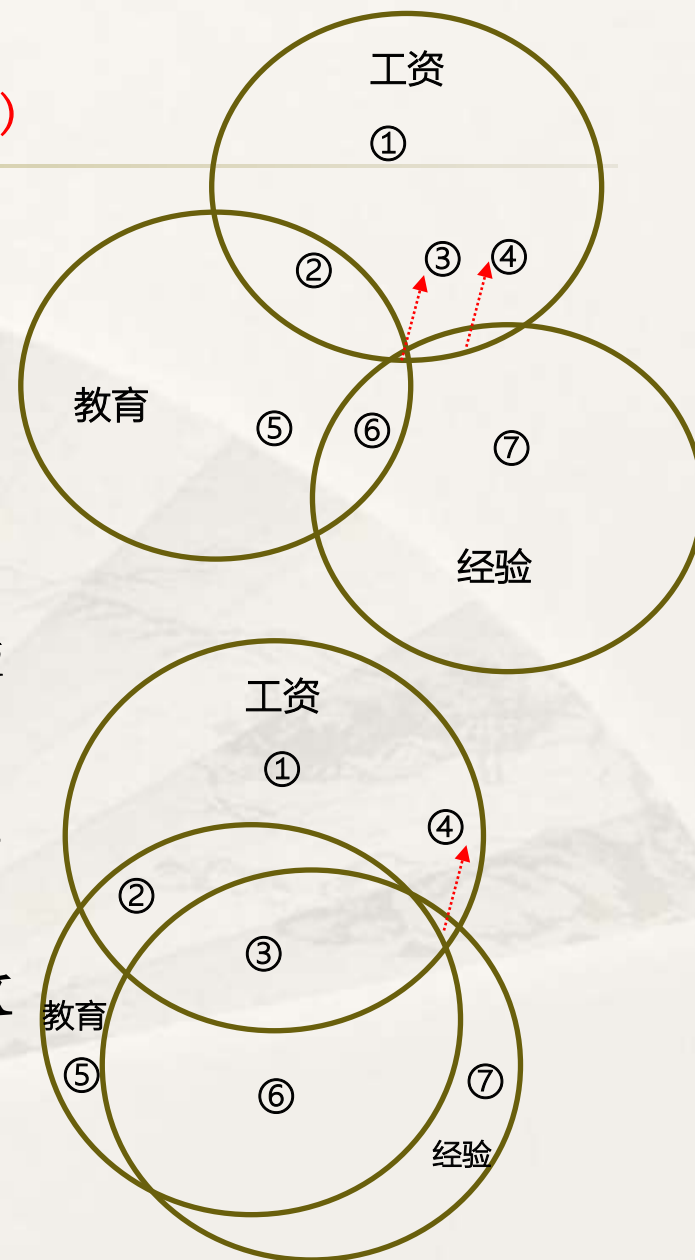
■ 多元模型 (过度设置模型: 无偏估计) 与一元模型 (简单模型: 有偏估计) 中的估计

■ 一元回归和多元回归: $\tilde{y} = \tilde{\beta}_0 + \tilde{\beta}_1 x_1$ $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2$.

■ 两个估计的关系: $\tilde{\beta}_1 = \hat{\beta}_1 + \hat{\beta}_2 \tilde{\delta}_1$ δ_1 是舍弃的变量 x_2 对自变量 x_1 的回归系数

■ 几种情况 (重点关注 $\tilde{\delta}_1$ 的影响):

- x_2 对 y 无显著 **净** 影响, 且与 x_1 弱相关 (如前例中的年级, 调节变量), 选择简单模型, 有偏但偏差近似为 0 且小方差
- x_2 对 y 无显著 **净** 影响, 但与 x_1 强相关 (偏差不近似为 0), 选择复杂模型 (无偏大方差), 简单模型估计有偏较大
 - 引入 x_2 的目的是获得 x_1 的无偏估计, 即使 x_2 的系数较小 (统计显著性和实际显著性) 也可考虑保留在模型中
 - 可能的交互效应 (调节效应) 显著



模型误设问题

- **模型设定不足** (under-specifying the model): 未包含**需被控制**的对 y 有显著线性影响的变量 (变量遗漏)

- 例: $\log(\text{wage}) = \beta_0 + \beta_1 \text{educ} + \beta_2 \text{exper} + u$ 遗漏了 exper

- 模型设定不足往往表现为对实际问题的某个方面未加以考虑, 这个方面通常需通过一组变量来反应

- 例: 美国棒球职业联赛运动员的年薪(y) (MLB1.txt)

$$\log(\text{salary}) = \beta_0 + \beta_1 \text{years} + \beta_2 \text{gamesyr} + u.$$

$$\log(\text{salary}) = \beta_0 + \beta_1 \text{years} + \beta_2 \text{gamesyr} + \beta_3 \text{bavg} + \beta_4 \text{hrunsyr} + \beta_5 \text{rbisyr} + u,$$

- 变量含义依次为: **加盟年数** (关注的自变量)、平均每年参加比赛的次数、击球率、本垒打次数、击球跑垒得分
- 判断是否需要考虑球技水平对年薪的影响作用

模型设定不足的问题

- 模型设定不足，遗漏变量的后果：
 - 若 u 中的遗漏变量与 x 相关，导致OLS估计量的一致性

$$\begin{aligned}\text{plim } \hat{\beta}_1 &= \beta_1 + \text{Cov}(x_1, u) / \text{Var}(x_1) \\ &= \beta_1, \text{ because } \text{Cov}(x_1, u) = 0\end{aligned}$$

- 若 u 中的遗漏变量与 x 相关，导致OLS估计量有偏
 - 通常需分析：有偏的方向、有偏的大小

模型设定不足的问题

- 合理模型为：

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + u, \quad \hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2$$

- 设置不足模型为 (如一元)：

$$\tilde{y} = \tilde{\beta}_0 + \tilde{\beta}_1 x_1$$

- 例：

$$wage = \beta_0 + \beta_1 educ + \beta_2 abil + u.$$

$$wage = \beta_0 + \beta_1 educ + v, \quad v = \beta_2 abil + u.$$

$$\tilde{\beta}_1 = \frac{\sum_{i=1}^n (x_{i1} - \bar{x}_1) y_i}{\sum_{i=1}^n (x_{i1} - \bar{x}_1)^2}$$

$$\tilde{\beta}_1 = \hat{\beta}_1 + \hat{\beta}_2 \tilde{\delta}_1$$

$$x_2 = \tilde{\delta}_0 + \tilde{\delta}_1 x_1$$

$$E(\tilde{\beta}_1) = \beta_1 + \beta_2 \tilde{\delta}_1$$

$$E(\tilde{\beta}_1) = \beta_1 + \beta_2 \frac{\sum_{i=1}^n (x_{i1} - \bar{x}_1) x_{i2}}{\sum_{i=1}^n (x_{i1} - \bar{x}_1)^2}.$$

$\tilde{\delta}_1$ 仅为 x 的函数

$$E(\tilde{\beta}_1) - \beta_1 = \beta_2 \tilde{\delta}_1$$

遗漏变量偏差
(Omitted variable bias)

模型设定不足的问题

- 遗漏变量偏差: $E(\tilde{\beta}_1) - \beta_1 = \beta_2 \tilde{\delta}_1$
 - $\beta_2=0$, 即 x_2 对 y 没有净影响, 遗漏变量偏差为0
 - $\delta_1=0$, 即 x_1 和 x_2 不相关时, 遗漏变量偏差为0
 - $\beta_2 \delta_1$ 不等于0时, 产生遗漏变量偏差

通常可凭经验
判断 β_2 的正负

	$\text{Corr}(x_1, x_2) > 0$	$\text{Corr}(x_1, x_2) < 0$
$\beta_2 > 0$	positive bias	negative bias
$\beta_2 < 0$	negative bias	positive bias

$E(\tilde{\beta}_1) > \beta_1$ $\longrightarrow$ 向上的偏误 (upward bias)
 $E(\tilde{\beta}_1) < \beta_1$ $\longrightarrow$ 向下的偏误 (downward bias)

- 例(3.6):教育回报率问题中, 估计不考虑EXPER的教育回报率(8.3%)的遗漏变量偏差

$$\log(wage) = \beta_0 + \beta_1 edu + u \quad \log(wage) = \beta_0 + \beta_1 edu + \beta_2 exper + u$$

. reg LNWAGE EDU

Source	SS	df	MS	Number of obs	=	526
Model	27.5606291	1	27.5606291	F(1, 524)	=	119.58
Residual	120.769121	524	.230475422	Prob > F	=	0.0000
				R-squared	=	0.1858
				Adj R-squared	=	0.1843
Total	148.32975	525	.282532857	Root MSE	=	.48008

LNWAGE	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]
EDUC	.0827444	.0075667	10.94	0.000	.0678796 .0976092
_cons	.5837727	.0973358	6.00	0.000	.3925563 .7749891

. reg LNWAGE EDU EXPER

Source	SS	df	MS	Number of obs	=	526
Model	36.9850401	2	18.4925201	F(2, 523)	=	86.86
Residual	111.34471	523	.212896195	Prob > F	=	0.0000
				R-squared	=	0.2493
				Adj R-squared	=	0.2465
Total	148.32975	525	.282532857	Root MSE	=	.46141

LNWAGE	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]
EDUC	.0979356	.0076224	12.85	0.000	.0829613 .1129099
EXPER	.0103469	.0015551	6.65	0.000	.0072919 .013402
_cons	.2168544	.108595	2.00	0.046	-.0035183 .4301904

. pwcorr EDU EXPER, sig

	EDUC	EXPER
EDUC	1.0000	
EXPER	-0.2995 0.0000	1.0000

$$E(\tilde{\beta}_1) - \beta_1 = \beta_2 \tilde{\delta}_1$$

- 未包含EXPER的教育回报率估计平均偏低。8.3%的估计存在向下偏误的可能性
- 遗漏变量无法测度时只能依据理论判断与自变量、因变量间的相关性正负
- 问: 包含EXPER模型的估计(9.7%)是否仍存在遗漏变量偏差?

$$\log(wage) = \beta_0 + \beta_1 edu + \beta_2 exper + \beta_3 abil + u$$

遗漏?

- 例:教育回报率问题 $\log(wage) = \beta_0 + \beta_1 edu + \beta_2 exper + \beta_3 abil + u$

- 合理模型为: $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + u,$

遗漏

- 误设(设置不足)模型为(多元模型): $\tilde{y} = \tilde{\beta}_0 + \tilde{\beta}_1 x_1 + \tilde{\beta}_2 x_2.$

- 因 x_1 与遗漏变量 x_3 相关, 误设模型的 $\tilde{\beta}_1$ 是有偏估计

$$E(\tilde{\beta}_1) = \beta_1 + \beta_3 \tilde{\delta}_1 \quad x_3 = \delta_0 + \tilde{\delta}_1 x_1 + \tilde{\delta}_2 x_2$$

- 变量 (x_1, x_2) 间的相关性使多元模型的偏差方向分析较为困难

- 粗略的分析方法: 假定 x_1 、 x_2 无关简化多元模型中遗漏变量偏差分析, 有:

$$E(\tilde{\beta}_1) = \beta_1 + \beta_3 \frac{\sum_{i=1}^n (x_{i1} - \bar{x}_1)x_{i3}}{\sum_{i=1}^n (x_{i1} - \bar{x}_1)^2}.$$

	$\text{Corr}(x_1, x_3) > 0$	$\text{Corr}(x_1, x_3) < 0$
$\beta_3 > 0$	positive bias	negative bias
$\beta_3 < 0$	negative bias	positive bias

- 示例结论: 9.7%存在高估的可能性

模型设定不足的诊断

- 例：美国棒球职业联赛运动员的年薪(y) (MLB1.txt)

$$\log(\text{salary}) = \beta_0 + \beta_1 \text{years} + \beta_2 \text{gamesyr} + u.$$

$$\log(\text{salary}) = \beta_0 + \beta_1 \text{years} + \beta_2 \text{gamesyr} + \beta_3 \text{bavg} + \beta_4 \text{hrunsyr} + \beta_5 \text{rbisyr} + u,$$

- 变量含义依次为：加盟年数、平均每年参加比赛的次数、击球率、本垒打次数、击球跑垒得分
- 判断是否需要考虑球技水平对年薪的影响作用
 - 第一个模型是否存在模型设定不足的问题
 - 在现有模型下，对一组变量能否进入模型进行诊断

■ 问：可否采用t检验？

■ 答：不恰当

$$H_0: \beta_3 = 0, \beta_4 = 0, \beta_5 = 0.$$

$$\log(\text{salary}) = \beta_0 + \beta_1 \text{years} + \beta_2 \text{gamesyr} + \beta_3 \text{bavg} + \beta_4 \text{hrunsyr} + \beta_5 \text{rbisyr} + u,$$

```
. reg LSALARY YEARS GAMESYR BAVG HRUNSYR RBISYR
```

Source	SS	df	MS	Number of obs	=	353
Model	308.989216	5	61.7978431	F(5, 347)	=	117.06
Residual	183.186321	347	.52791447	Prob > F	=	0.0000
				R-squared	=	0.6278
				Adj R-squared	=	0.6224
Total	492.175537	352	1.39822596	Root MSE	=	.72658

LSALARY	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
YEARS	.0688626	.0121145	5.68	0.000	.0450355	.0926898
GAMESYR	.0125521	.0026468	4.74	0.000	.0073464	.0177578
BAVG	.0009786	.0011035	0.89	0.376	-.0011918	.003149
HRUNSYR	.0144295	.016057	0.90	0.369	-.0171518	.0460107
RBISYR	.0107657	.007175	1.50	0.134	-.0033462	.0248776
_cons	11.19242	.2888228	38.75	0.000	10.62435	11.76048

两个或多个t统计量不显著的变量，联合起来可能十分显著！

•为什么？

```
lm(formula = LSALARY ~ ., data = Data)
```

Residuals:

Min	1Q	Median	3Q	Max
-3.02508	-0.45034	-0.04013	0.47014	2.68924

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1.119e+01	2.888e-01	38.752	< 2e-16 ***
YEARS	6.886e-02	1.211e-02	5.684	2.79e-08 ***
GAMESYR	1.255e-02	2.647e-03	4.742	3.09e-06 ***
BAVG	9.786e-04	1.104e-03	0.887	0.376
HRUNSYR	1.443e-02	1.606e-02	0.899	0.369
RBISYR	1.077e-02	7.175e-03	1.500	0.134

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.7266 on 347 degrees of freedom
Multiple R-squared: 0.6278, Adjusted R-squared: 0.6224
F-statistic: 117.1 on 5 and 347 DF, p-value: < 2.2e-16

k=6, 样本量N=353

模型设定不足的诊断

- 在现有模型下，对一组变量可否进入模型进行诊断，是一组参数显著性的检验，需整体检验
- 方法：
 - 拟合优度：依据引入一组变量后的拟合效果判断
 - 联合假设检验 (joint hypotheses test)：在模型参数加以约束的条件下，检验一组变量整体是否对因变量有线性影响 (适用于嵌套模型)

模型设定不足的诊断

- 决定系数 R^2 :

$$SST \equiv \sum_{i=1}^n (y_i - \bar{y})^2$$

$$SSE \equiv \sum_{i=1}^n (\hat{y}_i - \bar{y})^2$$

$$SSR \equiv \sum_{i=1}^n \hat{u}_i^2.$$

$$SST = SSE + SSR.$$

$$SSR/SST + SSE/SST = 1.$$

$$R^2 \equiv SSE/SST = 1 - SSR/SST,$$

(SSR: Residual Sum of Squares)

- 调整 R^2 (6.3节):

- 增加任一解释变量 x 都会使 R^2 增大。不可以 R^2 是否增大作为 x 对 y 有贡献的判断标准

- 调整为:

$$\begin{aligned}\bar{R}^2 &\equiv 1 - [SSR/(n - k - 1)]/[SST/(n - 1)] \\ &= 1 - \hat{\sigma}^2/[SST/(n - 1)],\end{aligned}$$

- 增加一个解释变量 x , SSR会下降, 但因 $n - k - 1$ 也下降, 因此均方误差可能下降也可能上升
- 有效的 x 应是使均方误差下降的变量 (SSR减少较大)

模型设定不足的诊断

- 联合假设检验 (4.5节)：在模型参数加以约束的条件下，检验一组变量整体是否对因变量有线性影响，适用于两个具有嵌套关系的模型

- 模型：
$$y = \beta_0 + \beta_1 x_1 + \dots + \beta_k x_k + u_i$$
 非约束模型

- 检验后 q 个解释变量是否应进入模型，原假设：

$$H_0: \beta_{k-q+1} = 0, \dots, \beta_k = 0$$

有多个排除性约束
(多重约束)

- 在约束条件下的模型为：

$$y = \beta_0 + \beta_1 x_1 + \dots + \beta_{k-q} x_{k-q} + u_i$$
 约束模型

- 约束模型和非约束模型存在嵌套关系，即前者是后者的特殊形式

模型设定不足的诊断

- 方法：F检验

$$y = \beta_0 + \beta_1 x_1 + \dots + \beta_k x_k + u_i \quad \text{非约束模型}$$

$$H_0: \beta_{k-q+1} = 0, \dots, \beta_k = 0$$

$$y = \beta_0 + \beta_1 x_1 + \dots + \beta_{k-q} x_{k-q} + u_i \quad \text{约束模型}$$

$$F \equiv \frac{(\text{SSR}_r - \text{SSR}_{ur})/q}{\text{SSR}_{ur}/(n - k - 1)}$$

约束模型的残差平方和

非约束模型的残差平方和

$$F \sim F_{q, n-k-1}$$

- F统计量非负，度量了残差平方和SSR从约束模型到非约束模型的相对减少程度。F充分大时应拒绝原假设，表示联合显著

■ F统计量与拟合优度 R^2 的关系

$$F \equiv \frac{(\text{SSR}_r - \text{SSR}_{ur})/q}{\text{SSR}_{ur}/(n - k - 1)}$$



$$\text{SSR}_r = \text{SST}(1 - R_r^2)$$

$$\text{SSR}_{ur} = \text{SST}(1 - R_{ur}^2)$$

$$F \equiv \frac{(R_{ur}^2 - R_r^2)/q}{(1 - R_{ur}^2)/(n - k - 1)}$$

回归方程的显著性检验：
F检验



$$\frac{R^2/k}{(1 - R^2)/(n - k - 1)}$$

$y = \beta_0 + u$ 约束模型 $R_r^2 = 0$

$$F = \frac{\text{SSE}/k}{\text{SSR}/(n - k - 1)}$$

总之，F检验可实现：回归方程的显著性检验；模型设定不足的诊断(针对嵌套模型)；

■ 例：美国棒球职业联赛运动员的年薪(y) (MLB1.txt)

```
> mbl_lm1<-update(mbl_lm,formula=~YEARS+GAMESYR) #仅包括两个变量的回归
> summary(mbl_lm1)
```

Call:

```
lm(formula = LSALARY ~ YEARS + GAMESYR, data = Data)
```

Residuals:

Min	1Q	Median	3Q	Max
-2.66858	-0.46412	-0.01176	0.49219	2.68829

$$\log(\text{salary}) = \beta_0 + \beta_1 \text{years} + \beta_2 \text{gamesyr} + \beta_3 \text{bavg} + \beta_4 \text{hrunsyr} + \beta_5 \text{rbisyr} + u,$$

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	11.223804	0.108312	103.625	< 2e-16 ***
YEARS	0.071318	0.012505	5.703	2.5e-08 ***
GAMESYR	0.020174	0.001343	15.023	< 2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.7527 on 350 degrees of freedom
Multiple R-squared: 0.5971, Adjusted R-squared: 0.5948
F-statistic: 259.3 on 2 and 350 DF, p-value: < 2.2e-16

```
> anova(mbl_lm,mbl_lm1)
```

Analysis of Variance Table

Model 1: LSALARY ~ YEARS + GAMESYR + BAVG + HRUNSYR + RBISYR

Model 2: LSALARY ~ YEARS + GAMESYR

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	347	183.19				
2	350	198.31	-3	-15.125	9.5503	4.474e-06 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

拒绝棒球技能对年薪没有显著影响的假设，不考虑棒球技能的模型存在模型设定不足问题

■ 例：美国棒球职业联赛运动员的年薪(y) (MLB1.xlsx)

```
. nestreg : regress LSALARY (YEARS GAMESYR) (BAVG HRUNSYR RBISYR)
```

Block 1: YEARS GAMESYR

Source	SS	df	MS	Number of obs	=	353
Model	293.864069	2	146.932034	F(2, 350)	=	259.32
Residual	198.311468	350	.566604195	Prob > F	=	0.0000
				R-squared	=	0.5971
				Adj R-squared	=	0.5948
Total	492.175537	352	1.39822596	Root MSE	=	.75273

LSALARY	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
YEARS	.071318	.012505	5.70	0.000	.0467236	.0959124
GAMESYR	.0201745	.0013429	15.02	0.000	.0175334	.0228156
_cons	11.2238	.108312	103.62	0.000	11.01078	11.43683

$$\log(\text{salary}) = \beta_0 + \beta_1 \text{years} + \beta_2 \text{gamesyr} + \beta_3 \text{bavg} + \beta_4 \text{hrunsyr} + \beta_5 \text{rbisyr} + u,$$

Block 2: BAVG HRUNSYR RBISYR

Source	SS	df	MS	Number of obs	=	353
Model	308.989216	5	61.7978431	F(5, 347)	=	117.06
Residual	183.186321	347	.52791447	Prob > F	=	0.0000
				R-squared	=	0.6278
				Adj R-squared	=	0.6224
Total	492.175537	352	1.39822596	Root MSE	=	.72658

LSALARY	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
YEARS	.0688626	.0121145	5.68	0.000	.0450355	.0926898
GAMESYR	.0125521	.0026468	4.74	0.000	.0073464	.0177578
BAVG	.0009786	.0011035	0.89	0.376	-.0011918	.003149
HRUNSYR	.0144295	.016057	0.90	0.369	-.0171518	.0460107
RBISYR	.0107657	.007175	1.50	0.134	-.0033462	.0248776
_cons	11.19242	.2888228	38.75	0.000	10.62435	11.76048

Block	F	Block Residual df	df	Pr > F	R2	Change in R2
1	259.32	2	350	0.0000	0.5971	
2	9.55	3	347	0.0000	0.6278	0.0307

拒绝棒球技能对年薪没有显著影响的假设，不考虑棒球技能的模型是存在模型设定不足问题的

```
. quietly regress LSALARY YEARS GAMESYR BAVG HRUNSYR RBISYR
```

```
. test BAVG HRUNSYR RBISYR
```

- (1) BAVG = 0
- (2) HRUNSYR = 0
- (3) RBISYR = 0

联合显著性检验

F(3, 347) = 9.55
Prob > F = 0.0000

模型设定不足的诊断

- 拉格朗日乘数 (Lagrange multiplier, LM) 统计量 (5.2节)
 - 大样本下，借助拟合优度构造检验统计量

- 目标：

$$y = \beta_0 + \beta_1 x_1 + \dots + \beta_k x_k + u_i$$

非约束模型

$$H_0: \beta_{k-q+1} = 0, \dots, \beta_k = 0$$

原假设

- 后 q 个自变量同时具有零总体参数

■ 建立约束模型：

$$y = \tilde{\beta}_0 + \tilde{\beta}_1 x_1 + \dots + \tilde{\beta}_{k-q} x_{k-q} + \tilde{u}$$

约束模型的
残差项

- 若原假设成立意味着约束模型的残差项与被排除的 q 个 x 近似无关
 - 因为： $\tilde{u} = \beta_{k-q+1} x_{k-q+1} + \dots + \beta_k x_k + v$ ，原假设成立时 $\tilde{u} = v$ ，根据残差性质 v 即 $\tilde{u}$ 与 $x_1, x_2 \dots x_k$ 无关
 - 通过 $\tilde{u}$ 与 $x_1, x_2 \dots x_k$ 的回归，判断可否推翻原假设

■ 基本步骤：

- 用残差对 k 个变量进行回归(辅助回归)： $\tilde{u} \text{ on } x_1, x_2, \dots, x_k$

- 计算辅助回归的拟合优度： R_u^2

若原假设成立，应接近0

- 构造： $LM = nR_u^2$ ，近似服从 q 个自由度卡方分布
- 若LM大于临界值，则应拒绝原假设

■ 例：美国棒球职业联赛运动员的年薪研究

```
Call:
lm(formula = mbl_lm1$residuals ~ . - LSALARY, data = Data)

Residuals:
    Min       1Q   Median       3Q      Max
-3.02508 -0.45034 -0.04013  0.47014  2.68924

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -0.0313857   0.2888228  -0.109   0.91353
YEARS        -0.0024553   0.0121145  -0.203   0.83951
GAMESYR      -0.0076224   0.0026468  -2.880   0.00423 **
BAVG          0.0009786   0.0011035   0.887   0.37579
HRUNSYR       0.0144295   0.0160570   0.899   0.36947
RBISYR        0.0107657   0.0071750   1.500   0.13441
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.7266 on 347 degrees of freedom
Multiple R-squared: 0.07627, Adjusted R-squared: 0.06296
F-statistic: 5.73 on 5 and 347 DF, p-value: 4.266e-05

> (LM<-0.07627*353) #LM统计量
[1] 26.92331
> 1-pchisq(q=LM,df=3) #计算概率P值
[1] 6.109378e-06
```

$$\log(\text{salary}) = \beta_0 + \beta_1 \text{years} + \beta_2 \text{gamesyr} + \beta_3 \text{bavg} + \beta_4 \text{hrunsyr} + \beta_5 \text{rbisyr} + u,$$

k=6, 样本量N=353

拒绝棒球技能对年薪没有显著影响的假设，不考虑棒球技能的模型存在模型设定不足问题

$$LM = nR_u^2$$

■ 例：美国棒球职业联赛运动员的年薪研究

```
. quietly reg LSALARY YEARS GAMESYR
. predict e1,residual
. reg e1 YEARS GAMESYR BAVG HRUNSYR RBISYR
```

Source	SS	df	MS	Number of obs	=	353
Model	15.1251471	5	3.02502942	F(5, 347)	=	5.73
Residual	183.186321	347	.527914469	Prob > F	=	0.0000
				R-squared	=	0.0763
				Adj R-squared	=	0.0630
Total	198.311468	352	.563384852	Root MSE	=	.72658

e1	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
YEARS	-.0024553	.0121145	-0.20	0.840	-.0262825	.0213718
GAMESYR	-.0076224	.0026468	-2.88	0.004	-.0128281	-.0024166
BAVG	.0009786	.0011035	0.89	0.376	-.0011918	.003149
HRUNSYR	.0144295	.016057	0.90	0.369	-.0171518	.0460107
RBISYR	.0107657	.007175	1.50	0.134	-.0033462	.0248776
_cons	-.0313857	.2888228	-0.11	0.914	-.5994495	.536678

```
. di "LM = " e(N)*e(r2)
LM = 26.923188

. di invchi2tail(3,0.05)
7.8147279
```

$$\log(\text{salary}) = \beta_0 + \beta_1 \text{years} + \beta_2 \text{gamesyr} + \beta_3 \text{bavg} + \beta_4 \text{hrunsyr} + \beta_5 \text{rbisyr} + u,$$

$$LM = nR_u^2$$

- LM大于临界值，则应拒绝原假设
 - 拒绝棒球技能对年薪没有显著影响的假设，不考虑棒球技能的模型是存在模型设定不足问题的

F检验的延展应用

- 一般线性假设检验
- 结构性变化检验

延展应用：一般线性假设检验

- 例：研究表明上世纪发达国家的教育回报率（排除其他因素影响）大约10%
 - 问：依WAGE1数据的估计(9.7%)能否支撑上述研究结论？
 - 研究目标：检验教育回报率的估计值与10%是否存在显著差异
 - 方法：一般线性假设检验(Linear Hypothesis Testing)

```
. reg LNWAGE EDU EXPER
```

Source	SS	df	MS	Number of obs	=	526
Model	36.9850401	2	18.4925201	F(2, 523)	=	86.86
Residual	111.34471	523	.212896195	Prob > F	=	0.0000
Total	148.32975	525	.282532857	R-squared	=	0.2493
				Adj R-squared	=	0.2465
				Root MSE	=	.46141

LNWAGE	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
EDUC	.0979356	.0076224	12.85	0.000	.0829613	.1129099
EXPER	.0103469	.0015551	6.65	0.000	.0072919	.013402
_cons	.2168544	.108595	2.00	0.046	.0035183	.4301904

延展应用：一般线性假设检验

- 一般线性假设检验 (4.5)：可用于对经验猜想的验证性分析
 - 思路：对前述F检验的再认识
 - 对比约束模型与非约束模型的残差，判断是否有显著差异。有显著差异表明不支持假设(猜想)
 - 例：
 - 非约束模型：
$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_4 + u,$$
 - 约束模型
 - 原假设：
$$H_0: \beta_1 = 1, \beta_2 = 0, \beta_3 = 0, \beta_4 = 0.$$
 - 约束模型：
$$y = \beta_0 + x_1 + u.$$
 - 估计模型：
$$y - x_1 = \beta_0 + u.$$

- 检验步骤：
 - 建立基于原假设的约束模型并估计
 - 例：

$$Y = \beta_0 + \beta_1 X_1 + \cdots + \beta_5 X_5 + \varepsilon, \quad H_0 : \beta_0 = \beta_1 = \beta_2 = 0$$

$$Y = \beta_3 X_3 + \beta_4 X_4 + \beta_5 X_5 + \varepsilon = \beta_0^* Z_0 + \beta_1^* Z_1 + \beta_2^* Z_2 + \varepsilon$$

$$\beta_0^* = \beta_3, \beta_1^* = \beta_4, \beta_2^* = \beta_5, Z_0 = X_3, Z_1 = X_4, Z_2 = X_5$$

约束OLS回归

- 构造F检验统计量

$$F \equiv \frac{(\text{SSR}_r - \text{SSR}_{ur})/q}{\text{SSR}_{ur}/(n - k - 1)}$$

- 若概率P值小于显著性水平，表明两个模型的残差有显著差异，拒绝原假设

•R实现中提出原假设:

■ 例:

$$Y = \beta_0 + \beta_1 X_1 + \cdots + \beta_5 X_5 + \varepsilon. \text{ Test } H_0 : \beta_0 = \beta_1 = \beta_2 = 0$$

$$\Leftrightarrow H_0 : C\beta = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \beta_3 \\ \beta_4 \\ \beta_5 \end{bmatrix} = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

给出约束式的左右项

■ 例:

$$\begin{aligned} Y &= \beta_0 + \beta_1 X_1 + \cdots + \beta_4 X_4 + \varepsilon \\ H_0 : \beta_1 - \beta_2 &= 0 \\ \dots\dots\dots 2\beta_3 - 3\beta_4 &= 0 \end{aligned} \Leftrightarrow H_0 : C\beta = \begin{bmatrix} 0 & 1 & -1 & 0 & 0 \\ 0 & 0 & 0 & 2 & -3 \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \beta_3 \\ \beta_4 \end{bmatrix} = \begin{bmatrix} \beta_1 - \beta_2 \\ 2\beta_3 - 3\beta_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

■ $C\beta$ 为 $q \times p$ 的阵 (q 为约束个数, p = 解释变量个数+1)

■ 教育回报率问题中: $q = 1, p = 3$

$$\log(wage) = \beta_0 + \beta_1 edu + \beta_2 exper + u$$

$$H_0 : \beta_1 = 0.10$$

$$C = (0, 1, 0); \beta = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{pmatrix}; C\beta = \beta_1 = 0.10$$

•例：教育回报率问题。结论：不能拒绝原假设，没充分证据拒绝10%的研究结论

$$H_0 : \beta_1 = 0.1$$

$$\log(\text{wage}) = \beta_0 + \beta_1 \text{educ} + \beta_2 \text{exper} + u = \beta_0 + 0.1 \text{educ} + \beta_2 \text{exper} + u$$

$$\log(\text{wage}) - 0.1 \text{educ} = \beta_0^* + \beta_1^* \text{exper} + u$$

```
> lgWAGE_lm2<-lm(log(WAGE)~EDUC+EXPER,data=Data)
> Data$z<-Data$EDUC*0.1
> Data$y<-log(Data$WAGE)-Data$z
> lgWAGE_tmp<-lm(y~EXPER,data=Data)
> deviance(lgWAGE_tmp)
[1] 111.3603
> deviance(lgWAGE_lm2)
[1] 111.3447
> (SS<-deviance(lgWAGE_tmp)-deviance(lgWAGE_lm2))
[1] 0.01561652
> (SS/1)/(deviance(lgWAGE_lm2)/(526-3)) # (SS/df)/无约束模型的均方误差
[1] 0.07335273
```

$$F \equiv \frac{(SSR_r - SSR_{ur})/q}{SSR_{ur}/(n - k - 1)}$$

```
> linearHypothesis(lgWAGE_lm2,c("EDUC=0.1"))
```

Linear hypothesis test

Hypothesis:

EDUC = 0.1

Model 1: restricted model

Model 2: log(WAGE) ~ EDUC + EXPER

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	524	111.36				
2	523	111.34	1	0.015617	0.0734	0.7866

$$H_0 : \beta_1 = 0.10$$

$$C = (0,1,0); \beta = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{pmatrix}; C\beta = \beta_1 = 0.10$$

```
> linearHypothesis(lgWAGE_lm2,c(0,1,0),rhs=0.1)
```

Linear hypothesis test

Hypothesis:

EDUC = 0.1

Model 1: restricted model

Model 2: log(WAGE) ~ EDUC + EXPER

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	524	111.36				
2	523	111.34	1	0.015617	0.0734	0.7866

•例：教育回报率问题

带约束的回归

$$H_0: \beta_1 = 0.1$$

$$\log(wage) = \beta_0 + \beta_1 educ + \beta_2 exper + u = \beta_0 + 0.1educ + \beta_2 exper + u$$

$$\log(wage) - 0.1educ = \beta_0^* + \beta_1^* exper + u$$

```
. constraint def 1 EDU=0.1 //设置第1个约束条件
```

```
. cnsreg LNWAGE EDU EXPER,c(1)
```

```
Constrained linear regression      Number of obs    =      526
                                   Root MSE          =      0.4610
```

```
( 1)  EDUC = .1
```

LNWAGE	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
EDUC	.1 (constrained)					
EXPER	.0104731	.0014824	7.06	0.000	.0075609	.0133853
_cons	.1887726	.0322553	5.85	0.000	.125407	.2521382

```
. predict e1,residual
```

$$F \equiv \frac{(SSR_r - SSR_{ur})/q}{SSR_{ur}/(n - k - 1)}$$

```
. reg LNWAGE EDU EXPER //非约束模型
```

Source	SS	df	MS	Number of obs	=	526
Model	36.9850401	2	18.4925201	F(2, 523)	=	86.86
Residual	111.34471	523	.212896195	Prob > F	=	0.0000
Total	148.32975	525	.282532857	R-squared	=	0.2493
				Adj R-squared	=	0.2465
				Root MSE	=	.46141

LNWAGE	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
EDUC	.0979356	.0076224	12.85	0.000	.0829613	.1129099
EXPER	.0103469	.0015551	6.65	0.000	.0072919	.013402
_cons	.2168544	.108595	2.00	0.046	.0035183	.4301904

```
. predict e2,residual
```

```
. egen rss1=sum(e1^2)
```

```
. egen rss2=sum(e2^2)
```

```
. di "F = " ((rss1-rss2)/1)/(rss2/(e(N)-2-1))
```

```
F = .07335674
```

```
. di invFtail(1,(e(N)-3),0.05)
```

```
3.8593004
```

```
. test EDU=0.1
```

```
( 1)  EDUC = .1
```

```
F( 1, 523) = 0.07
Prob > F = 0.7866
```

•结论：不能拒绝原假设，没有充分证据拒绝10%的研究结论

延展应用：结构性变化研究

- 例：累计GPA的影响因素分析(基于GPA3.txt中SPRING=1 (N=366)的数据集)

$$cumgpa = \beta_0 + \beta_1 sat + \beta_2 hsperc + \beta_3 tothrs + u,$$

- 累计GPA, Sat成绩, hsperc高中成绩排名, tothrs大学课程总学分

- 目标：

- 分析男女生的累计GPA是否有显著差异(其他不变条件下)
- 影响累计GPA的因素在性别间是否有差异(性别的调节作用)

$$cumgpa = \beta_0 + \delta_0 female + \beta_1 sat + \delta_1 female \cdot sat + \beta_2 hsperc + \delta_2 female \cdot hsperc + \beta_3 tothrs + \delta_3 female \cdot tothrs + u.$$

- 原假设： $H_0: \delta_0 = 0, \delta_1 = 0, \delta_2 = 0, \delta_3 = 0.$

- 本质：两总体的结构性变化研究

- 若无法推翻原假设，表示两总体的回归函数没有显著差异，无结构性变化

■ 问：是否可以利用t检验？

```
lm(formula = CUMGPA ~ FEMALE * (SAT + HSPERC + TOTHR), data = Data)

Residuals:
    Min       1Q   Median       3Q      Max
-1.51370 -0.28645 -0.02306  0.27555  1.24760

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)   1.4808117   0.2073336   7.142 5.17e-12 ***
FEMALE1      -0.3534862   0.4105293  -0.861 0.38979
SAT           0.0010516   0.0001811   5.807 1.40e-08 ***
HSPERC        -0.0084516   0.0013704  -6.167 1.88e-09 ***
TOTHR         0.0023441   0.0008624   2.718 0.00688 **
FEMALE1:SAT    0.0007506   0.0003852   1.949 0.05211 .
FEMALE1:HSPERC -0.0005498   0.0031617  -0.174 0.86206
FEMALE1:TOTHR  -0.0001158   0.0016277  -0.071 0.94331
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.4678 on 358 degrees of freedom
Multiple R-squared:  0.4059,    Adjusted R-squared:  0.3943
F-statistic: 34.95 on 7 and 358 DF,  p-value: < 2.2e-16
```

尽管各系数检验均不显著，
但需联合检验

■ 方法：对约束和非约束模型做差异的联合检验(F检验)

■ 非约束模型：

$$\text{cumgpa} = \beta_0 + \delta_0 \text{female} + \beta_1 \text{sat} + \delta_1 \text{female} \cdot \text{sat} + \beta_2 \text{hsperc} + \delta_2 \text{female} \cdot \text{hsperc} + \beta_3 \text{tothrs} + \delta_3 \text{female} \cdot \text{tothrs} + u.$$

■ 约束模型：

$$\text{cumgpa} = \beta_0 + \beta_1 \text{sat} + \beta_2 \text{hsperc} + \beta_3 \text{tothrs} + u,$$

■ 例：累计GPA研究

```
> deviance(ungpa)      非约束模型
[1] 78.35451
> gpa<-lm(CUMGPA~SAT+HSPERC+TOTHRs,data=Data)  ###约束模型
> summary(gpa)

Call:
lm(formula = CUMGPA ~ SAT + HSPERC + TOTHRs, data = Data)

Residuals:
    Min       1Q   Median       3Q      Max
-1.59576 -0.31150 -0.01042  0.29369  1.35348

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  1.4908498  0.1836782   8.117 7.57e-15 ***
SAT           0.0011850  0.0001648   7.191 3.71e-12 ***
HSPERC       -0.0099569  0.0012446  -8.000 1.70e-14 ***
TOTHRs        0.0023429  0.0007554   3.102 0.00208 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.486 on 362 degrees of freedom
Multiple R-squared:  0.3516,    Adjusted R-squared:  0.3463
F-statistic: 65.44 on 3 and 362 DF,  p-value: < 2.2e-16

> deviance(gpa)
[1] 85.51507
> (deviance(gpa)-deviance(ungpa))/deviance(ungpa)*(366-7-1)/4
[1] 8.179111
```

$$\text{cumgpa} = \beta_0 + \delta_0 \text{female} + \beta_1 \text{sat} + \delta_1 \text{female} \cdot \text{sat} + \beta_2 \text{hsperc} + \delta_2 \text{female} \cdot \text{hsperc} + \beta_3 \text{tothrs} + \delta_3 \text{female} \cdot \text{tothrs} + u.$$

$$\text{cumgpa} = \beta_0 + \beta_1 \text{sat} + \beta_2 \text{hsperc} + \beta_3 \text{tothrs} + u,$$

$$F \equiv \frac{(\text{SSR}_r - \text{SSR}_{ur})/q}{\text{SSR}_{ur}/(n - k - 1)}$$

结论：不同性别的GPA回归模型存在显著差异，两总体存在结构性变动

■ 例：累计GPA研究

$$cumgpa = \beta_0 + \beta_1 sat + \beta_2 hspers + \beta_3 tothrs + u,$$

```
. nestreg : regress CUMGPA (HSPERC TOTHR SAT) (FEMALE FSAT FHSPERC FTOTHR) if SPRING==1
```

Block 1: HSPERC TOTHR SAT

Source	SS	df	MS	Number of obs	=	366
Model	46.3786191	3	15.4595397	F(3, 362)	=	65.44
Residual	85.5150683	362	.23622947	Prob > F	=	0.0000
				R-squared	=	0.3516
				Adj R-squared	=	0.3463
Total	131.893687	365	.361352568	Root MSE	=	.48603

CUMGPA	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
HSPERC	-.0099569	.0012446	-8.00	0.000	-.0124044	-.0075094
TOTHR	.0023429	.0007554	3.10	0.002	.0008574	.0038285
SAT	.001185	.0001648	7.19	0.000	.0008609	.001509
_cons	1.49085	.1836782	8.12	0.000	1.12964	1.85206

$$cumgpa = \beta_0 + \delta_0 female + \beta_1 sat + \delta_1 female \cdot sat + \beta_2 hspers + \delta_2 female \cdot hspers + \beta_3 tothrs + \delta_3 female \cdot tothrs + u.$$

Block 2: FEMALE FSAT FHSPERC FTOTHR

Source	SS	df	MS	Number of obs	=	366
Model	53.53918	7	7.64845428	F(7, 358)	=	34.95
Residual	78.3545074	358	.218867339	Prob > F	=	0.0000
				R-squared	=	0.4059
				Adj R-squared	=	0.3943
Total	131.893687	365	.361352568	Root MSE	=	.46783

CUMGPA	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
HSPERC	-.0084516	.0013704	-6.17	0.000	-.0111465	-.0057566
TOTHR	.0023441	.0008624	2.72	0.007	.0006482	.0040401
SAT	.0010516	.0001811	5.81	0.000	.0006955	.0014078
FEMALE	-.3534862	.4105293	-0.86	0.390	-1.160838	.4538659
FSAT	.0007506	.0003852	1.95	0.052	-6.88e-06	.0015081
FHSPERC	-.0005498	.0031617	-0.17	0.862	-.0067676	.0056681
FTOTHR	-.0001158	.0016277	-0.07	0.943	-.0033169	.0030852
_cons	1.480812	.2073336	7.14	0.000	1.073067	1.888557

Block	Block F	Block df	Residual df	Pr > F	R2	Change in R2
1	65.44	3	362	0.0000	0.3516	
2	8.18	4	358	0.0000	0.4059	0.0543

$$F \equiv \frac{(SSR_r - SSR_{ur})/q}{SSR_{ur}/(n - k - 1)}$$

结论：不同性别的GPA回归模型存在显著差异，两总体存在结构性变动

- 一般研究认为：不同总体回归模型的截距项不同, 仅需对交互项进行联合检验
 - 例：累计GPA研究

$$\text{cumgpa} = \beta_0 + \delta_0 \text{female} + \beta_1 \text{sat} + \delta_1 \text{female} \cdot \text{sat} + \beta_2 \text{hsperc} + \delta_2 \text{female} \cdot \text{hsperc} + \beta_3 \text{tothrs} + \delta_3 \text{female} \cdot \text{tothrs} + u.$$

```
> gpa<-lm(CUMGPA~FEMALE+SAT+HSPERC+TOTHRS,data=Data) ###约束模
> summary(gpa)
```

$$H_0: \delta_0 = 0, \delta_1 = 0, \delta_2 = 0, \delta_3 = 0.$$

```
Call:
lm(formula = CUMGPA ~ FEMALE + SAT + HSPERC + TOTHRS, data = Da
```

Residuals:

Min	1Q	Median	3Q	Max
-1.55122	-0.29553	-0.01821	0.29436	1.18113

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1.3285406	0.1798275	7.388	1.04e-12 ***
FEMALE1	0.3100975	0.0586128	5.291	2.12e-07 ***
SAT	0.0012144	0.0001591	7.635	2.04e-13 ***
HSPERC	-0.0084413	0.0012343	-6.839	3.43e-11 ***
TOTHRS	0.0024638	0.0007291	3.379	0.000806 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.4689 on 361 degrees of freedom
 Multiple R-squared: 0.3983, Adjusted R-squared: 0.3916
 F-statistic: 59.74 on 4 and 361 DF, p-value: < 2.2e-16

```
> deviance(gpa)
```

```
[1] 79.36167
```

```
> (deviance(gpa)-deviance(ungpa))/deviance(ungpa)*(366-7-1)/3
```

```
[1] 1.533898
```

结论：无法拒绝原假设。不同性别总体的GPA结构变动仅源于不同性别在累计GPA上的差异，其他变量对累计GPA的影响效应不存在性别上的显著差异(不存在显著的性别调节作用)

- 一般研究认为：不同总体回归模型的截距项不同。仅需对交互项进行联合检验

■ 例：累计GPA研究

$$\text{cumgpa} = \beta_0 + \delta_0 \text{female} + \beta_1 \text{sat} + \delta_1 \text{female} \cdot \text{sat} + \beta_2 \text{hsperc} + \delta_2 \text{female} \cdot \text{hsperc} + \beta_3 \text{tothrs} + \delta_3 \text{female} \cdot \text{tothrs} + u.$$

```
. nestreg : regress CUMGPA (HSPERC TOTHRs SAT FEMALE) (FSAT FHSPERC FTOTHR) if SPRING==1
```

$$H_0: \delta_0 = 0, \delta_1 = 0, \delta_2 = 0, \delta_3 = 0.$$

Block 1: HSPERC TOTHRs SAT FEMALE

Source	SS	df	MS	Number of obs	=	366
Model	52.5320197	4	13.1330049	F(4, 361)	=	59.74
Residual	79.3616677	361	.219838415	Prob > F	=	0.0000
				R-squared	=	0.3983
				Adj R-squared	=	0.3916
Total	131.893687	365	.361352568	Root MSE	=	.46887

CUMGPA	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
HSPERC	-.0084413	.0012343	-6.84	0.000	-.0108687	-.0060139
TOTHRs	.0024638	.0007291	3.38	0.001	.00103	.0038976
SAT	.0012144	.0001591	7.63	0.000	.0009016	.0015272
FEMALE	.3100975	.0586128	5.29	0.000	.1948321	.4253629
_cons	1.328541	.1798275	7.39	0.000	.9748995	1.682182

Block 2: FSAT FHSPERC FTOTHRs

Source	SS	df	MS	Number of obs	=	366
Model	53.53918	7	7.64845428	F(7, 358)	=	34.95
Residual	78.3545074	358	.218867339	Prob > F	=	0.0000
				R-squared	=	0.4059
				Adj R-squared	=	0.3943
Total	131.893687	365	.361352568	Root MSE	=	.46783

CUMGPA	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
HSPERC	-.0084516	.0013704	-6.17	0.000	-.0111465	-.0057566
TOTHRs	.0023441	.0008624	2.72	0.007	.0006482	.0040401
SAT	.0010516	.0001811	5.81	0.000	.0006955	.0014078
FEMALE	-.3534862	.4105293	-0.86	0.390	-1.160838	.4538659
FSAT	.0007506	.0003852	1.95	0.052	-6.88e-06	.0015081
FHSPERC	-.0005498	.0031617	-0.17	0.862	-.0067676	.0056681
FTOTHRs	-.0001158	.0016277	-0.07	0.943	-.0033169	.0030852
_cons	1.480812	.2073336	7.14	0.000	1.073067	1.888557

Block	Block F	Block df	Residual df	Pr > F	R2	Change in R2
1	59.74	4	361	0.0000	0.3983	
2	1.53	3	358	0.2054	0.4059	0.0076

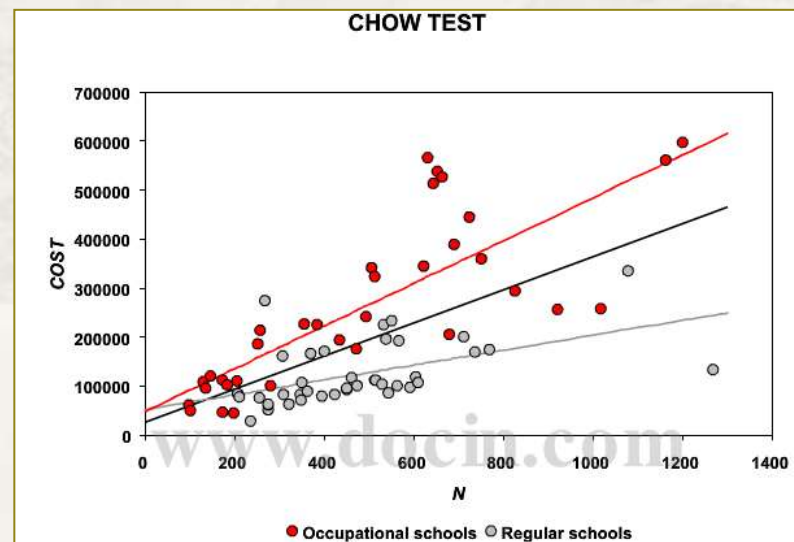
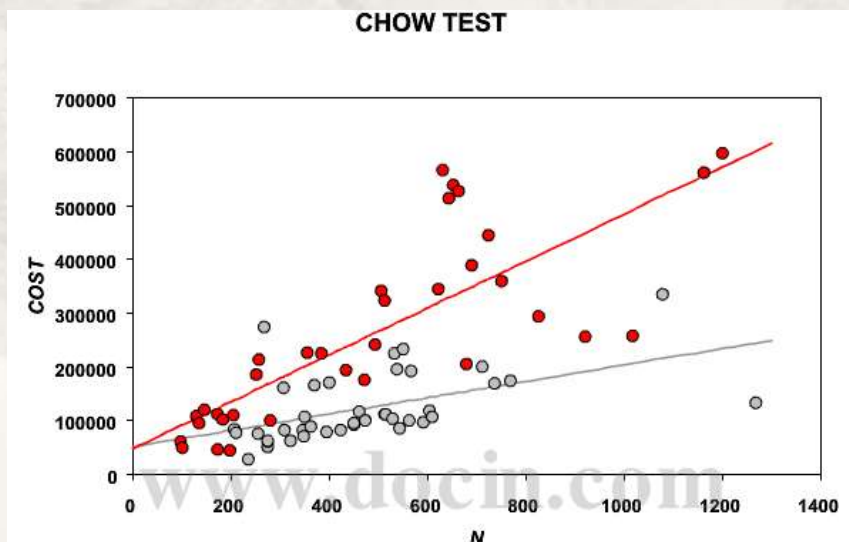
- 结论：无法拒绝原假设。不同性别总体的GPA结构变动仅源于不同性别在累计GPA上的差异，其他变量对累计GPA的影响效应不存在性别上的显著差异(不存在显著的性别调节作用)
- 引入虚拟自变量要求更大的样本量！！

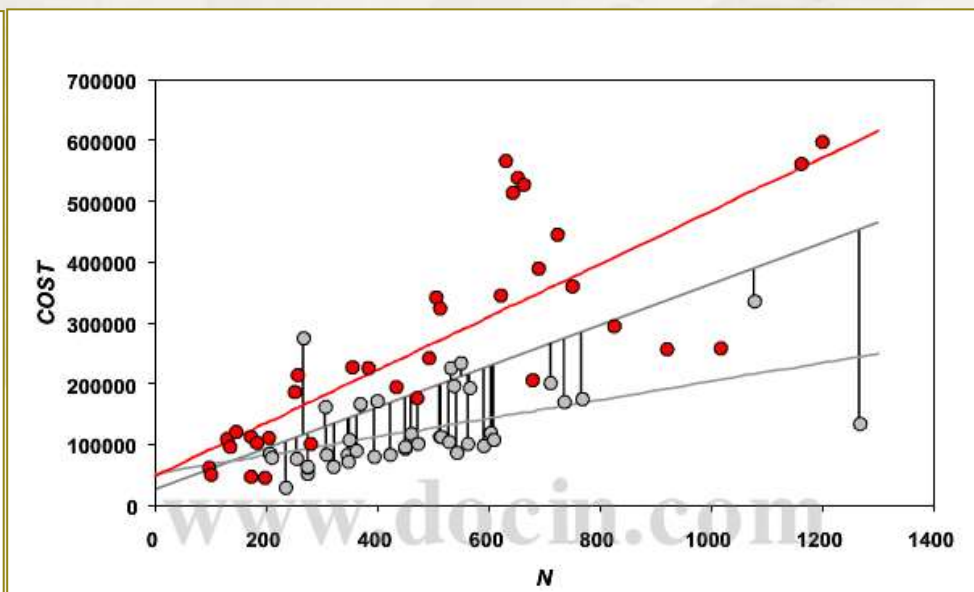
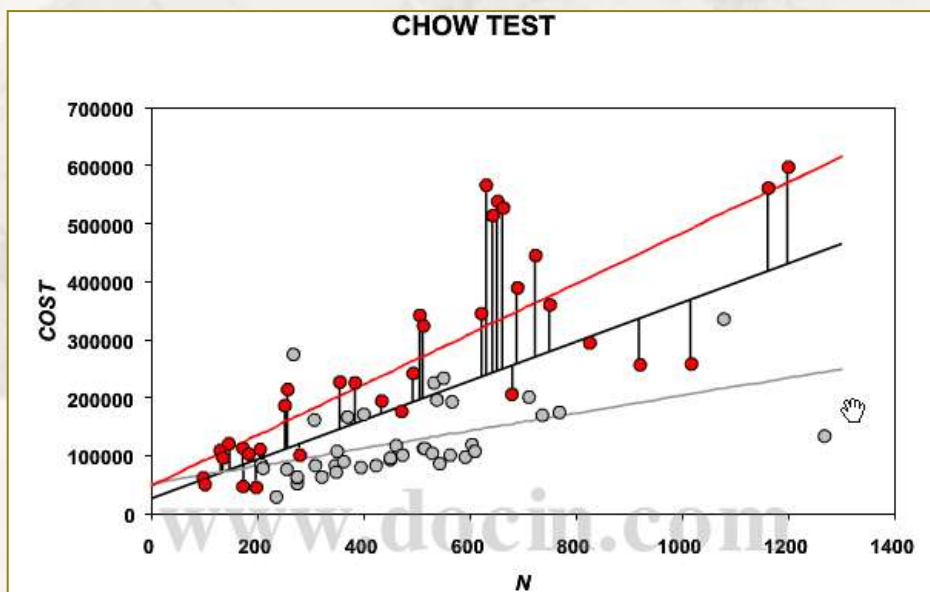
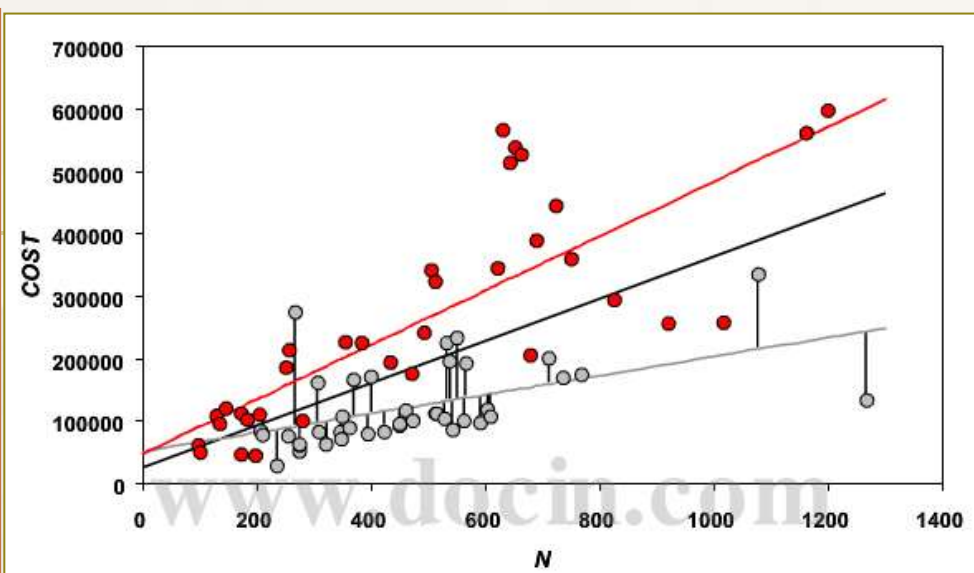
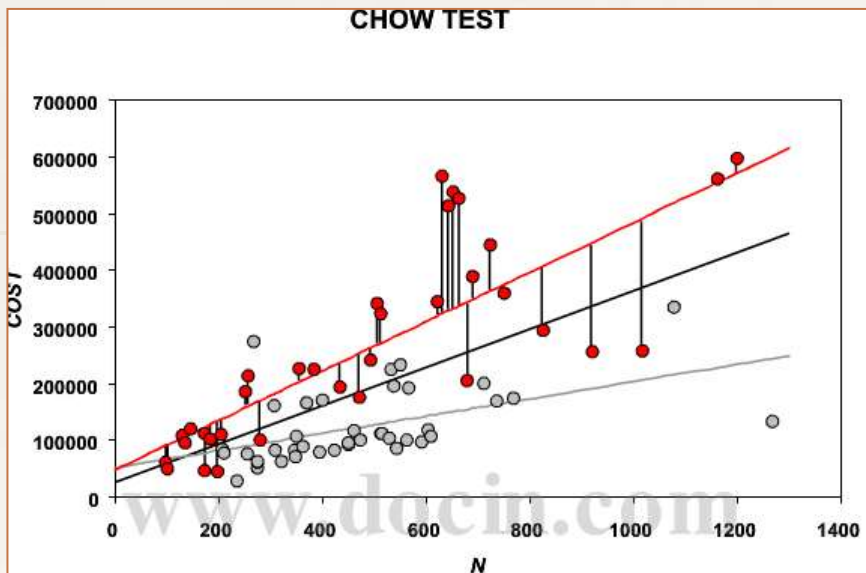
结构性变化研究：邹至庄(CHOW)检验

- 目的：检验不同总体的回归函数是否相同，可用于结构性变化、政策效果评价等方面(7.4)

- 例：
$$\text{cumgpa} = \beta_0 + \delta_0 \text{female} + \beta_1 \text{sat} + \delta_1 \text{female} \cdot \text{sat} + \beta_2 \text{hsperc} + \delta_2 \text{female} \cdot \text{hsperc} + \beta_3 \text{tothrs} + \delta_3 \text{female} \cdot \text{tothrs} + u.$$

$$H_0: \delta_0 = 0, \delta_1 = 0, \delta_2 = 0, \delta_3 = 0.$$





看残差平方和变化是否显著

延展应用：邹至庄检验

- 基本步骤：不采用虚拟自变量形式

- 分样本建立多组模型(约束模型)：

- 模型：

$$y = \beta_{g,0} + \beta_{g,1}x_1 + \beta_{g,2}x_2 + \dots + \beta_{g,k}x_k + u,$$

- g 代表不同组(如1,2), 残差自由度为 $n_g - k - 1$

分组建立模型意味默认结构性差异，本质为**非约束模型的残差**

- 计算多组模型的残差平方和之和： $SSR_{ur} = SSR_1 + SSR_2$.

- 自由度为： $n - 2(k + 1)$

- 对所有观测建立上述模型，残差平方和SSR

- 构造邹至庄统计量：

$$F = \frac{[SSR - (SSR_1 + SSR_2)]}{SSR_1 + SSR_2} \cdot \frac{[n - 2(k + 1)]}{k + 1} \rightarrow \text{自由度：} n - k - 1 - [n - 2k - 2] = k + 1$$

- 服从 $(k+1, n-2(k+1))$ 个自由度的F分布

例：累计GPA研究

```
> ungpa<-lm(CUMGPA~FEMALE*(SAT+HSPERC+TOTHR),data=Data) ##非约束模型
> summary(ungpa)

Call:
lm(formula = CUMGPA ~ FEMALE * (SAT + HSPERC + TOTHR), data = Data)

Residuals:
    Min       1Q   Median       3Q      Max
-1.51370 -0.28645 -0.02306  0.27555  1.24760

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)   1.4808117   0.2073336   7.142 5.17e-12 ***
FEMALE1      -0.3534862   0.4105293  -0.861  0.38979
SAT           0.0010516   0.0001811   5.807 1.40e-08 ***
HSPERC       -0.0084516   0.0013704  -6.167 1.88e-09 ***
TOTHR        0.0023441   0.0008624   2.718  0.00688 **
FEMALE1:SAT   0.0007506   0.0003852   1.949  0.05211 .
FEMALE1:HSPERC -0.0005498  0.0031617  -0.174  0.86206
FEMALE1:TOTHR -0.0001158  0.0016277  -0.071  0.94331
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.4678 on 358 degrees of freedom
Multiple R-squared:  0.4059,    Adjusted R-squared:  0.3943
F-statistic: 34.95 on 7 and 358 DF,  p-value: < 2.2e-16
```

- 这里通过虚拟自变量方式实现（建立非约束模型）
- 应分性别分别建立两个约束模型

```
> deviance(ungpa)
[1] 78.35451
> gpa<-lm(CUMGPA~SAT+HSPERC+TOTHR,data=Data) ###约束模型
> deviance(gpa)
[1] 85.51507
> (deviance(gpa)-deviance(ungpa))/deviance(ungpa)*(366-2*(3+1))/(3+1) #CHOW统计量
[1] 8.179111
```

- 结构性变化研究：依次检验指定区间内的每个观测点，是否为潜在变点 (breakpoint)
 - 可用于时间序列研究中，分析变点前(包括变点)和后是否出现了结构性变动

1978-2006年(共29年)
y人均GDP, c人均收入

library("strucchange")

```
> Data<-read.table(file="breakpoint.txt",header=TRUE,sep=",")
> Result<-Fstats(c~y,data=Data)
> Result$Fstats # #CHOW统计量*(k+1)
```

Time Series:

Start = c(0, 5)

End = c(0, 26)

Frequency = 29

```
[1] 2.865754 3.715315 4.719967 6.006425 7.388731 9.104111 11.319680 13.252603 15.724253 19.837422
[11] 27.116722 40.326735 72.514498 175.623357 389.524725 481.630816 681.143478 1021.250206 1202.021743 1162.170177
[21] 690.115060 328.219481
```

```
> Result$Fstats/(1+1)
```

Time Series:

Start = c(0, 5)

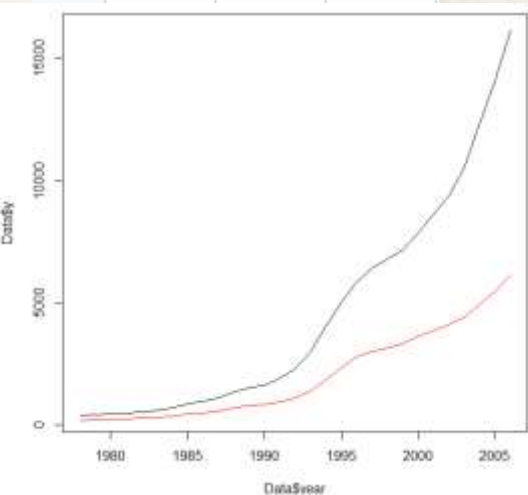
End = c(0, 26)

Frequency = 29

```
[1] 1.432877 1.857657 2.359983 3.003212 3.694366 4.552056 5.659840 6.626301 7.862127 9.918711
[11] 13.558361 20.163367 36.257249 87.811678 194.762362 240.815408 340.571739 510.625103 601.010872 581.085089
[21] 345.057530 164.109741
```

```
> plot(Result$Fstats/(1+1),type="o")
```

$$F = \frac{[SSR - (SSR_1 + SSR_2)]}{SSR_1 + SSR_2} \cdot \frac{[n - 2(k + 1)]}{k + 1}$$

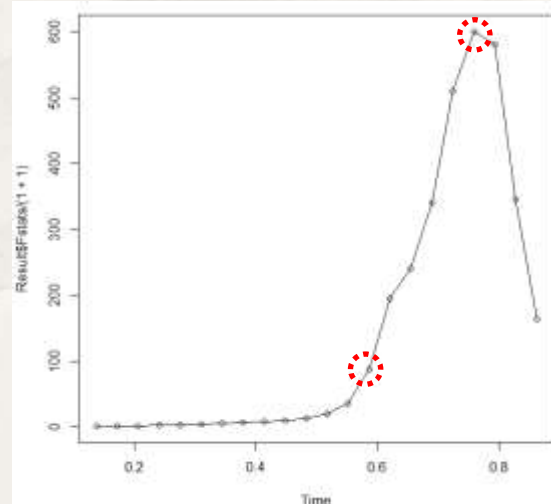


消费结构性变动研究:在1982-2003年中找变点

- 1978-1982, 1983-2006
- 1978-1983, 1984-2006.....
- 1978-2003, 2004-2006 (n>p+1)

- 1996年开始出现较大变动
- 2000年邹至庄统计量达到最大值

SSR对应全样本,SSR1和SSR2
分别对应两时期样本



模型误设问题

- 函数形式误设 (functional form misspecification)

- 注意体现在：模型中遗漏了解释变量的非线性函数

$$\log(\text{wage}) = \beta_0 + \beta_1 \text{educ} + \beta_2 \text{exper} + \beta_3 \text{exper}^2 + u \quad \text{遗漏了 } \text{exper}^2$$

$$\log(\text{wage}) = \beta_0 + \beta_1 \text{educ} + \beta_2 \text{exper} + \beta_3 \text{exper}^2 + \beta_4 \text{female} + \beta_5 \text{female} \cdot \text{educ} + u, \quad \text{遗漏了交互项}$$

- 问：F检验能否对此进行检验？答：能
 - 不足：事先决定应引入怎样的自变量函数形式(应是二次或是三次或是对数等)较为困难
 - 多个解释变量引入导致的自由度问题

模型误设问题

- RESET (regression specification error test) 法：拉姆齐的回归设定误差检验法 (9.1节)：诊断是否遗漏了解释变量的非线性函数

- 若模型设定正确则任何解释变量的非线性关系进入模型应不显著
- 处理方法：

$$y = \beta_0 + \beta_1 x_1 + \dots + \beta_k x_k + u$$

$$y = \beta_0 + \beta_1 x_1 + \dots + \beta_k x_k + \delta_1 \hat{y}^2 + \delta_2 \hat{y}^3 + error$$

- 如果原模型是正确的，则无法推翻原假设：

$$H_0: \delta_1 = 0, \delta_2 = 0$$

自变量的非线性函数

- 两模型具有嵌套形式，可用F检验或LM检验

■ RESET法示例 (例:9.2, HPRICE1): 房屋价格(阁楼平米、总平米、卧室间数)

$$price = \beta_0 + \beta_1 lotsize + \beta_2 sqrft + \beta_3 bdrms + u.$$

```
> library("lmtest")
> resettest(hprice_lm)
```

•R默认引入拟合值
的二阶和三阶

$$y = \beta_0 + \beta_1 x_1 + \dots + \beta_k x_k + \delta_1 \hat{y}^2 + \delta_2 \hat{y}^3 + error$$

RESET test

```
data: hprice_lm
RESET = 4.6682, df1 = 2, df2 = 82, p-value = 0.01202
```

5%显著性水平下, 拒绝原假设, 存在函数形式误设(遗漏了解释变量的非线性函数)

```
. reg PRICE LOTSIZE SQRFT BDRMS
```

Source	SS	df	MS	Number of obs	=	88
Model	617130.702	3	205710.234	F(3, 84)	=	57.46
Residual	300723.806	84	3580.04531	Prob > F	=	0.0000
				R-squared	=	0.6724
				Adj R-squared	=	0.6607
Total	917854.508	87	10550.0518	Root MSE	=	59.833

PRICE	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]
LOTSIZE	.0020677	.0006421	3.22	0.002	.0007908 .0033446
SQRFT	.1227782	.0132374	9.28	0.000	.0964541 .1491022
BDRMS	13.85252	9.010145	1.54	0.128	-4.06514 31.77018
_cons	-21.77031	29.47504	-0.74	0.462	-80.38466 36.84404

```
. estat ovtest
```

Ramsey RESET test using powers of the fitted values of PRICE
Ho: model has no omitted variables

F(3, 81) = 4.26
Prob > F = 0.0076

- 非线性项包括拟合值的2次、3次和4次项
- 5%显著性水平下存在函数形式误设(遗漏了解释变量的非线性函数)

■ 利用RESET法检验其他函数形式误设问题

■ 目标:

■ 检验模型应设置为怎样的形式更可取

- 拟合优度检验可行?
- 采用RESET方法检验

RESET test

```
data: hprice_lm  
RESET = 4.6682, df1 = 2, df2 = 82, p-value = 0.01202
```

- 对比两个模型的RESET检验结果: 哪个模型的概率P值越大, 哪个模型更可取

• 5%的显著性水平上不存在函数形式误设。对数模型更可取

$$\begin{aligned} \text{price} &= \beta_0 + \beta_1 \text{lotsize} + \beta_2 \text{sqrft} + \beta_3 \text{bdrms} + u. \\ \text{lprice} &= \beta_0 + \beta_1 \text{llotsize} + \beta_2 \text{lsqrft} + \beta_3 \text{bdrms} + u. \end{aligned}$$

```
> summary(hprice_lm<-lm(LPRICE~.-SQRFT-LOTSIZE-PRICE,data=Data))
```

Call:

```
lm(formula = LPRICE ~ . - SQRFT - LOTSIZE - PRICE, data = Data)
```

Residuals:

Min	1Q	Median	3Q	Max
-0.68422	-0.09178	-0.01584	0.11213	0.66899

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-1.29704	0.65128	-1.992	0.0497 *
BDRMS	0.03696	0.02753	1.342	0.1831
LSQRFT	0.70023	0.09287	7.540	5.01e-11 ***
LLOTSIZE	0.16797	0.03828	4.388	3.31e-05 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.1846 on 84 degrees of freedom

Multiple R-squared: 0.643, Adjusted R-squared: 0.6302

F-statistic: 50.42 on 3 and 84 DF, p-value: < 2.2e-16

```
> resettest(hprice_lm)
```

RESET test

```
data: hprice_lm
```

```
RESET = 2.563, df1 = 2, df2 = 82, p-value = 0.08308
```

- 利用RESET法检验其他函数形式误设问题
 - 目标：检验模型应设置为怎样的形式更可取

```
. estat ovtest
```

Ramsey RESET test using powers of the fitted values of PRICE

Ho: model has no omitted variables

F(3, 81) = 4.26

Prob > F = 0.0076

$$price = \beta_0 + \beta_1 lotsize + \beta_2 sqrft + \beta_3 bdrms + u.$$

$$lprice = \beta_0 + \beta_1 llotsize + \beta_2 lsqrft + \beta_3 bdrms + u.$$

```
. reg LPRICE LLOTSIZE LSQRFT BDRMS
```

Source	SS	df	MS	Number of obs	=	88
Model	5.15503927	3	1.71834642	F(3, 84)	=	50.42
Residual	2.86256322	84	.034078134	Prob > F	=	0.0000
				R-squared	=	0.6430
				Adj R-squared	=	0.6302
Total	8.01760248	87	.09215635	Root MSE	=	.1846

LPRICE	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
LLOTSIZE	.1679667	.0382812	4.39	0.000	.0918404	.2440929
LSQRFT	.7002324	.0928653	7.54	0.000	.5155597	.8849051
BDRMS	.0369583	.0275313	1.34	0.183	-.0177907	.0917074
_cons	-1.297041	.6512836	-1.99	0.050	-2.59219	-.0018924

```
. estat ovtest
```

Ramsey RESET test using powers of the fitted values of LPRICE

Ho: model has no omitted variables

F(3, 81) = 2.45

Prob > F = 0.0692

5%的显著性水平上不存在函数形式误设。对比两个模型的RESET检验结果，对数模型更可取

模型误设问题

- 戴维森-麦金农 (Davidson-MacKinnon) 检验 (9.1节)

- 例：模型应设置为以下哪种形式

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + u$$

$$y = \beta_0 + \beta_1 \log(x_1) + \beta_2 \log(x_2) + u.$$

- 两个模型是非嵌套关系

- 基本思想：若某个模型正确，则另一个模型的拟合值在正确模型中应不显著

- 例：若对数模型正确，则水平模型的拟合值 $\hat{y}$ 在对数模型中应不显著

$$y = \beta_0 + \beta_1 \log(x_1) + \beta_2 \log(x_2) + \theta_1 \hat{y} + error;$$

$\hat{y}$ 的t检验应不显著

■ 戴维森-麦金农检验示例

■ 前例:

$$price = \beta_0 + \beta_1 lotsize + \beta_2 sqrft + \beta_3 bdrms + u.$$

$$lprice = \beta_0 + \beta_1 llotsize + \beta_2 lsqrft + \beta_3 bdrms + u.$$

- 建立水平模型
- 建立对数模型(包括水平模型的拟合值)
- t检验不显著,对数模型相对合理

不足: 即使t检验不显著, 也不能证明对数模型的形式就一定正确, 可能还有其他形式

```
. reg PRICE LOTSIZE SQRFT BDRMS
```

Source	SS	df	MS	Number of obs	=	88
Model	617130.702	3	205710.234	F(3, 84)	=	57.46
Residual	300723.806	84	3580.04531	Prob > F	=	0.0000
				R-squared	=	0.6724
				Adj R-squared	=	0.6607
Total	917854.508	87	10550.0518	Root MSE	=	59.833

PRICE	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
LOTSIZE	.0020677	.0006421	3.22	0.002	.0007908	.0033446
SQRFT	.1227782	.0132374	9.28	0.000	.0964541	.1491022
BDRMS	13.85252	9.010145	1.54	0.128	-4.06514	31.77018
_cons	-21.77031	29.47504	-0.74	0.462	-80.38466	36.84404

```
. predict yhat
(option xb assumed; fitted values)
```

```
. reg LPRICE LLOTSIZE LSQRFT BDRMS yhat
```

Source	SS	df	MS	Number of obs	=	88
Model	5.17094682	4	1.29273671	F(4, 83)	=	37.69
Residual	2.84665566	83	.034297056	Prob > F	=	0.0000
				R-squared	=	0.6449
				Adj R-squared	=	0.6278
Total	8.01760248	87	.09215635	Root MSE	=	.18519

LPRICE	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
LLOTSIZE	.1371915	.0593031	2.31	0.023	.0192401	.2551429
LSQRFT	.4892995	.3234294	1.51	0.134	-.1539886	1.132588
BDRMS	.0223228	.0349951	0.64	0.525	-.047281	.0919267
yhat	.0008336	.001224	0.68	0.498	-.0016009	.0032681
_cons	.3818477	2.550296	0.15	0.881	-4.690588	5.454283

模型误设问题

■ RainBow检验 (Utt 1982)

- 核心思想：若函数形式误设，则它可能对“中心”样本有较好拟合，但对“尾部”样本拟合不理想。
 - “中心”样本：
 - 计算各观测与中心点的马氏距离按升序排序，默认抽取前50%的观测
 - 或者，依某解释变量排序默认抽前50%观测
- 基于中心样本和全体样本的两个模型^{模型}的残差，检验差异是否显著。若差异显著则认为存在函数形式误设
- 检验统计量：
 - 原假设：两个模型的残存无显著差异
 - 这里中心样本量= m ，且为一元模型 $k = 1$

The sum of squares of error due to the lack of fit is defined as:

$$SSE_{LOF} = SSE_{FULL} - SSE_{CENTRAL}.$$

Hence

$$F = \frac{SSE_{LOF} / (n - m)}{SSE_{CENTRAL} / (m - 2)} \stackrel{H_0}{\sim} F(n - m, m - 2).$$

•例：房价模型的RAINBOW检验

$$price = \beta_0 + \beta_1 lotsize + \beta_2 sqrft + \beta_3 bdrms + u.$$

$$lprice = \beta_0 + \beta_1 llotsize + \beta_2 lsqrft + \beta_3 bdrms + u.$$

```
> hprice_lm<-lm(PRICE~.-LSQRFT-LLOTSIZE-LPRICE,data=Data)
> raintest(hprice_lm,fraction=0.5,order.by="mahalanobis",data=Data)
```

Rainbow test

```
data: hprice_lm
Rain = 2.1389, df1 = 44, df2 = 40, p-value = 0.008221
```

```
> raintest(hprice_lm, fraction=0.5,order.by=~SQRFT,data=Data)
```

Rainbow test

```
data: hprice_lm
Rain = 3.469, df1 = 44, df2 = 40, p-value = 6.042e-05
```

```
> raintest(hprice_lm, fraction=0.5,order.by=~BDRMS,data=Data)
```

Rainbow test

```
data: hprice_lm
Rain = 1.9601, df1 = 44, df2 = 40, p-value = 0.0166
```

```
> hprice_lm<-lm(LPRICE~.-SQRFT-LLOTSIZE-PRICE,data=Data)
> raintest(hprice_lm,fraction=0.5,order.by="mahalanobis",data=Data)
```

Rainbow test

```
data: hprice_lm
Rain = 1.2273, df1 = 44, df2 = 40, p-value = 0.257
```

```
Data<-read.table(file="hprice1.txt",header=TRUE,sep="")
library("lmtest")
```

- 水平模型存在函数形式误设
- 可用于对比解释变量对模型形式的影响(平米数较卧室间数影响效果大)
- 对数模型不存在函数形式误设